

Dynamic-Reference Cumulative Prospect Policy Gradients: Debiased Estimation, Minimax Limits, and Online Portfolio Tracking

Anonymous Authors

1 Introduction

Portfolio advice is inherently sequential and path dependent. An investor who reaches the same current wealth after a rally and drawdown may evaluate the next risky allocation very differently from an investor who reaches that wealth after a loss and recovery. In human-facing financial systems, this distinction matters because a policy that is attractive under expected return can be rejected once recent gains, losses, and reference levels are taken into account. Cumulative prospect theory (CPT) provides a canonical description of such behavior through reference dependence, asymmetric valuation of gains and losses, and nonlinear probability weighting [1, 2]. These features make CPT particularly relevant to sequential portfolio choice, but they also turn the learning objective into a nonlinear functional of the entire trajectory-outcome distribution rather than an additive one-period reward.

A growing literature has brought this distributional objective into reinforcement learning. Early work formulated prediction and control under CPT and established stochastic optimization procedures for prospect-theoretic objectives [3, 4]. Most closely related to this paper, Lepel and Barakat [5]—the current v5 version of the core CPT policy-gradient reference—derive a first-order policy-gradient theorem for finite-horizon CPT-RL, construct an order-statistics Monte Carlo estimator, and prove convergence and sample-complexity guarantees for first-order stationarity. This establishes a principled foundation for CPT policy optimization. Its reference point, however, enters as a fixed element of the CPT criterion rather than an endogenous state that is updated by the realized wealth path.

Behavioral evidence and recent portfolio models point in the opposite direction. Reference points adapt after realized gains and losses, with systematic asymmetry in the speed of adjustment [6, 7]; adaptive and path-dependent references can materially change realization and portfolio decisions [8–10]. The simplest example already changes the optimization problem. Starting from normalized wealth 1, consider a rise to 1.4 followed by a fall to 1.2, versus a fall to 0.8 followed by a recovery to 1.2. The terminal wealth is identical, but an upward-adjusted reference can place the first path in the loss domain while a lower reference places the second in the gain domain. Portfolio-level evidence that gain–loss framing affects the disposition effect reinforces the importance of this state dependence [11], while recent Chinese retail-investor evidence shows that disposition behavior also varies with information and investor characteristics [12].

Once the reference point is made endogenous, two difficulties that are separate under a fixed target become coupled. First, the CPT gradient depends on the derivative of a probability-weighting function evaluated at an unknown outcome distribution. Finite-rank plug-in evaluation therefore introduces a nonlinear estimation problem before the policy can be updated. Second, the target itself moves: market conditions generate exogenous variation, while the policy changes wealth, wealth changes the reference point, and the reference point changes the next CPT objective. Convergence to a single stationary point is no longer the appropriate criterion. The relevant question is whether a

feasible first-order policy can track a sequence of moving stationary conditions, and how its tracking error decomposes into target variation and statistical estimation error.

This paper develops *Dynamic-Reference Cumulative Prospect Policy Gradient* (DRCPT-PG) for this setting. The portfolio policy is a factor-parametrized Dirichlet distribution, which produces long-only simplex-valued allocations while keeping the parameter dimension tied to interpretable characteristics rather than the number of stocks. The reference state follows an asymmetric update after realized wealth changes. For learning, a centered leave-one-out score statistic is combined with an independent, untruncated randomized full/half correction. The resulting gradient output is exactly unbiased for the frozen regularized CPT target, has finite variance and finite expected trajectory cost, admits a studentized central limit theorem, and attains the fixed-dimensional minimax trajectory-budget rate. The online layer then uses a normalized projected update and evaluates tracking through the projected residual and dynamic local regret (DLR).

The analysis is organized around three linked questions. The first is statistical: can the nonlinear CPT policy gradient be estimated without finite-sample bias while retaining usable variance, inference, and budget guarantees? The second is dynamic: under what variation and simulation-error conditions does online updating keep the projected first-order residual small as the market and reference state evolve? The third is behavioral and empirical: does dynamic reference formation produce the path dependence and trading asymmetries predicted by behavioral finance, and can the resulting policy be evaluated coherently in real A-share portfolios, held-out investor behavior, and heterogeneous retail-agent interactions? This separation lets each layer of evidence test a distinct part of the argument rather than using portfolio returns as a proxy for estimator correctness or using behavioral fit as a proxy for optimization quality.

The theoretical results answer the first two questions. We derive the regularized CPT score-gradient identity; prove the finite-sample bias order of the centered leave-one-out base; construct an exactly unbiased estimator with a finite M^{-1} variance bound, asymptotic normality, and a minimax-optimal $\Theta(B^{-1})$ fixed-dimensional trajectory-budget rate; and establish a bound on the average expected squared projected residual for the online algorithm. Dynamic local regret is bounded by the cumulative squared projected residual, and a local error-bound condition converts the same residual control into distance to the moving stationary set. When objective variation is sublinear and the sampling and simulation errors vanish in Cesàro average, the expected average DLR vanishes.

The empirical program mirrors this logic. Path experiments first isolate the causal role of reference history in portfolio ranking. Fixed-target experiments then evaluate the estimator independently of online state propagation. Continuous slow-variation experiments test online tracking and dense sensitivity sweeps map the region over which the tracking advantage persists. The external-validity layers move from controlled mechanisms to real-market and behavioral evaluation: real A-share backtests compare economic performance under common trading constraints; investor-behavior tests use held-out decisions to assess whether asymmetric dynamic references improve the explanation and prediction of disposition behavior; and heterogeneous retail-agent interactions examine recommendation adoption and satisfaction across distinct risk profiles. This progression connects the theorem-level objects—gradient error, projected residual, and DLR—to market and behavioral outcomes without collapsing them into a single metric.

Contributions.

1. **Endogenous-reference CPT policy optimization.** We formulate a finite-horizon portfolio objective in which the psychological reference point is part of the evolving state rather than a fixed hyperparameter. The model yields explicit path dependence and a decomposition of

objective variation into non-reference and reference components; fixed-reference CPT-PG is recovered as a special case.

2. **Debiased gradient estimation with statistical guarantees.** We establish the regularized CPT score-gradient identity and construct a centered leave-one-out estimator with an independent randomized raw correction. The estimator is exactly unbiased, has finite expected cost and finite variance, obeys a multivariate central limit theorem, and attains the fixed-dimensional minimax rate under the same expected complete-trajectory budget.
3. **Online first-order tracking for a moving nonconvex target.** We analyze a normalized projected update through the projected residual. The resulting bound separates objective variation, replication error, and frozen-simulator error; it implies a DLR bound and, under an optional local error-bound condition, tracking of the moving stationary set.
4. **A layered empirical design from mechanism to deployment.** The experiments separately test reference-path dependence, estimator accuracy, continuous online tracking, and sensitivity, and extend the same framework to real A-share portfolio performance, held-out investor-behavior prediction, and heterogeneous retail-agent interaction. This design links behavioral interpretation to statistical and optimization guarantees while keeping each empirical claim tied to its corresponding estimand.

Organization. [Section 2](#) positions the paper relative to CPT optimization, adaptive-reference finance, nonstationary nonconvex learning, statistical debiasing, and human-facing financial agents. [Section 3](#) defines the dynamic-reference portfolio objective, projected tracking criterion, and the assumptions needed for the statistical and online analyses. [Section 4](#) presents the debiased gradient estimator and DRCPT-PG update. [Section 5](#) gives the estimator, minimax, projected-residual, DLR, and stationary-set results. [Section 6](#) organizes the mechanism, estimator, tracking, robustness, market, behavioral, and heterogeneous-agent evidence. [Section 7](#) synthesizes the implications and scope; proofs are collected in the appendix.

2 Related Work and Positioning

CPT optimization and reinforcement learning. Prospect theory and CPT replace expected-utility evaluation by a reference-dependent value transformation and nonlinear probability weighting [1, 2]. In sequential decision making, Prashanth et al. [3] showed that a CPT criterion must be treated as a functional of the return distribution rather than a transformed immediate reward, and Jie et al. [4] developed stochastic-optimization machinery for this class of objectives. The closest reference is Lepel and Barakat [5], whose v5/TMLR formulation derives a first-order CPT policy-gradient theorem, an order-statistics Monte Carlo estimator, and convergence and sample-complexity guarantees for approximate first-order stationarity. In parallel, recent optimization work has developed finite-scenario and portfolio-specific algorithms for static CPT objectives, including ADMM-based decision optimization [13], convex-analytic portfolio procedures [14], and a recent behavioral portfolio formulation that combines CPT with a flexible investment horizon [15]. Our work keeps the first-order CPT policy-gradient perspective but changes two structural elements: the reference becomes an endogenous state generated by realized wealth, and the learning objective therefore becomes a moving, path-dependent sequence. This requires both exact finite-sample debiasing of the frozen-target gradient and a tracking analysis for the resulting moving objective.

Adaptive references, path dependence, and investor behavior. Reference adaptation has long been documented in security-trading experiments, including asymmetric updating after equal-sized gains and losses [6, 7]. Mathematical-finance models then showed that adaptive references can amplify realization behavior [8], while path-dependent reference points make current decisions depend on the route by which wealth arrived at its present level [9]. More recent portfolio analysis embeds a dynamic reference directly in multi-period loss-averse allocation and finds materially different policies and disposition-effect predictions [10]. Complementary evidence shows that the disposition effect depends on portfolio-wide gain/loss frames [11], investor horizon and experience [16], and the information environment of Chinese retail investors [12]. These studies establish that the relevant behavioral state is neither fixed nor purely asset-local. They do not, however, provide an online CPT policy-gradient procedure whose parameter update and first-order tracking guarantees explicitly accommodate reference drift. That is the point at which the behavioral-reference literature and the optimization problem studied here meet.

Nonstationary and nonconvex online optimization. For nonconvex objectives, global comparator regret is generally too strong to characterize first-order learning. Local-regret criteria instead measure whether recent gradients or gradient mappings are small [17]. Aydore et al. [18] introduced dynamic local regret for nonconvex forecasting with time-smoothed gradients, and Hallak et al. [19] extended local-regret ideas to stochastic constrained problems through a proximal-gradient mapping. In nonstationary RL, variation budgets similarly quantify when a changing environment remains learnable [20]. Recent projected-gradient analysis sharpens first-order stationarity guarantees for constrained nonconvex and stochastic smooth optimization [21]. Our setting combines these ingredients but introduces a distinct source of variation: part of the objective drift is generated by the learner’s own realized wealth and reference dynamics. We therefore work with a normalized projected residual, derive DLR from that residual, and isolate objective variation and simulator error in the online bound.

Debiasing, inference, and information limits. The statistical difficulty in a CPT gradient is generated by a nonlinear functional of an unknown survival distribution. The order-statistics approach of Lepel and Barakat [5] provides a practical first-order estimator and consistency guarantees. Our estimator targets a different finite-sample property: exact unbiasedness for the regularized frozen-target gradient under a complete-trajectory oracle. Its independent randomized correction follows the general principle of unbiased randomized telescoping [22], while the leave-one-out expansion uses uniform empirical-distribution control of the type underlying the Dvoretzky–Kiefer–Wolfowitz inequality [23]. The central-limit and minimax layers connect the estimator to classical asymptotic and statistical-decision theory [24, 25]. The resulting contribution is therefore not merely a numerical bias correction: it specifies the mean, variance, limiting distribution, expected trajectory cost, and fixed-dimensional information rate of the gradient oracle used by the online algorithm.

Portfolio learning and A-share evaluation. The portfolio policy combines factor information with a simplex-valued action. Dirichlet factor policies provide a natural precedent for reinforced portfolio allocation because they generate valid long-only weights while linking concentration parameters to firm characteristics [26]. For the empirical market layer, recent work emphasizes both the time variation of financial-RL environments [27] and the importance of market-specific factor structure in Chinese A-shares [28]. These considerations motivate separating the controlled semi-synthetic environment used for theorem-aligned E2–E4 from the real A-share backtest used to evaluate return, risk, turnover, and drawdown under actual chronological market data.

Human-facing financial agents and heterogeneous risk preferences. Recent AI research increasingly treats behavioral preferences as part of the system objective rather than as evaluation noise. Prospect-theoretic optimization has been used to construct human-aware alignment losses for language models [29], and recent work directly evaluates whether LLMs represent persona-specific economic risk preferences [30]. In multi-agent learning, Lalmohammed et al. [31] embed CPT transformations into heterogeneous agents and show that risk profiles can materially change strategic behavior. Financial-advice studies add an interaction layer: personalized generative-AI advisors can influence investment choices, but preference elicitation, advice quality, satisfaction, and trust need not move together [32], while investor resistance to robo-advisors is shaped by both functional and psychological barriers [33]. These findings motivate the final behavioral layer of our design: held-out investor decisions are used to test the explanatory value of asymmetric dynamic references, and heterogeneous simulated retail agents are used to study recommendation adoption and satisfaction separately from portfolio performance.

3 Problem Setup

3.1 Nonstationary episodic portfolio process

Episodes are indexed by $k = 1, \dots, K$, begin at dates t_k , and have a fixed horizon $h \geq 1$. For consecutive episodes, $t_{k+1} = t_k + h$. At calendar time $u \in \{t_k, \dots, t_k + h - 1\}$,

$$s_u = (m_u, X_u, r_u, \omega_{u-1}), \quad \omega_u \in \Delta_N := \left\{ \omega \in \mathbb{R}_+^{N+1} : \sum_{i=0}^N \omega_i = 1 \right\}, \quad (1)$$

where asset 0 is cash, $X_u > 0$ is normalized wealth, and $r_u \geq 0$ is the reference wealth. For return vector R_{u+1} with $R_{0,u+1} = 0$ and proportional transaction-cost coefficient $c_{tc} \geq 0$,

$$X_{u+1} = X_u \left(1 + \omega_u^\top R_{u+1} - c_{tc} \|\omega_u - \omega_{u-1}\|_1 \right), \quad X_1 = 1. \quad (2)$$

The conditional market transition law may vary with calendar time. Conditional on the observed history and action, it has no explicit dependence on the policy parameter.

3.2 Dynamic reference and terminal relative outcome

For fixed $\eta_+, \eta_- \in [0, 1)$,

$$r_{u+1} = \Phi(r_u, X_{u+1}) = \begin{cases} (1 - \eta_+)r_u + \eta_+ X_{u+1}, & X_{u+1} \geq r_u, \\ (1 - \eta_-)r_u + \eta_- X_{u+1}, & X_{u+1} < r_u. \end{cases} \quad (3)$$

Let $\mathcal{F}_k$ be the information available at the start of episode k , and set

$$r_k := r_{t_k}, \quad z_k := (m_{t_k}, X_{t_k}, \omega_{t_k-1}). \quad (4)$$

All Monte Carlo learning trajectories for episode k are generated conditionally on the frozen context $(\mathcal{F}_k, z_k, r_k)$. Each trajectory τ carries its own wealth and reference recursions and produces

$$Y_k(\tau) = X_{t_k+h}(\tau) - r_{t_k+h}(\tau). \quad (5)$$

3.3 Dirichlet portfolio policy

For asset $i = 0, \dots, N$, let $q_{i,u} \in \mathbb{R}^d$ be a bounded feature vector. For $\theta \in \Theta \subset \mathbb{R}^d$ define

$$a_{i,u}(\theta) = \exp(\theta^\top q_{i,u}), \quad \tilde{\omega}_u \sim \pi_\theta(\cdot | s_u) = \text{Dirichlet}(a_{0,u}(\theta), \dots, a_{N,u}(\theta)). \quad (6)$$

A parameter-independent feasibility map $\mathbf{M}_u : \Delta_N \rightarrow \Delta_N$ and a fixed trade fraction $\chi \in (0, 1]$ generate the executed portfolio

$$\omega_u = (1 - \chi)\omega_{u-1} + \chi \mathbf{M}_u(\tilde{\omega}_u). \quad (7)$$

The realized path and all learning trajectories use the same stochastic actor (6) and the same map (7). The augmented trajectory records $\tilde{\omega}_u$, so its score is

$$G_{k,\theta}(\tau) = \sum_{u=t_k}^{t_k+h-1} \sum_{i=0}^N a_{i,u} q_{i,u} \left\{ \log \tilde{\omega}_{i,u} - \psi(a_{i,u}) + \psi \left(\sum_{j=0}^N a_{j,u} \right) \right\}, \quad (8)$$

where ψ is the digamma function.

3.4 Regularized CPT objective

For $x \geq 0$ let

$$u_{\alpha, \varepsilon_v}(x) = (x^2 + \varepsilon_v^2)^{\alpha/2} - \varepsilon_v^\alpha, \quad v_+(y) = u_{\alpha_+, \varepsilon_v}(y^+), \quad v_-(y) = \lambda u_{\alpha_-, \varepsilon_v}(y^-), \quad (9)$$

with $0 < \alpha_+, \alpha_- \leq 1$, $\lambda > 1$, and $\varepsilon_v > 0$. Let $w_\pm$ be strictly increasing probability-weighting functions and, for fixed $\varepsilon \in (0, 1/2)$, define the endpoint-normalized weights

$$w_\pm^\varepsilon(p) = \frac{w_\pm(\varepsilon + (1 - 2\varepsilon)p) - w_\pm(\varepsilon)}{w_\pm(1 - \varepsilon) - w_\pm(\varepsilon)}, \quad 0 \leq p \leq 1. \quad (10)$$

Thus $w_\pm^\varepsilon(0) = 0$ and $w_\pm^\varepsilon(1) = 1$.

For a frozen episode environment/model P and context (z, r) , let P_θ denote the induced augmented trajectory law under π_θ . Write

$$S_{P,z,r,\theta}^\pm(y) = P_\theta\{v_\pm(Y) > y\}.$$

On the bounded analysis domain, $v_\pm(Y) \in [0, B_v]$. Define

$$J(P, z, r; \theta) = \int_0^{B_v} w_+^\varepsilon(S_{P,z,r,\theta}^+(y)) dy - \int_0^{B_v} w_-^\varepsilon(S_{P,z,r,\theta}^-(y)) dy. \quad (11)$$

Let P_k denote the true conditional episode- k model and $\tilde{P}_k$ its frozen simulator model. Both are fixed when differentiating with respect to θ ; their induced trajectory laws depend on θ through the policy. Then

$$J_k(\theta, r_k) := J(P_k, z_k, r_k; \theta), \quad \tilde{J}_k(\theta, r_k) := J(\tilde{P}_k, z_k, r_k; \theta), \quad (12)$$

with gradients

$$g_k(\theta) := \nabla_\theta J_k(\theta, r_k), \quad \tilde{g}_k(\theta) := \nabla_\theta \tilde{J}_k(\theta, r_k). \quad (13)$$

3.5 Projected residual and dynamic local regret

Fix a normalization floor $\vartheta > 0$ and a step size $\gamma > 0$. Define

$$d(g) = \frac{g}{\max\{\|g\|_2, \vartheta\}}, \quad (14)$$

and the population projected residual

$$Q_k(\theta) = \frac{\Pi_{\Theta}\{\theta + \gamma d(g_k(\theta))\} - \theta}{\gamma}. \quad (15)$$

The constrained first-order stationary set is

$$S_k = \{\theta \in \Theta : Q_k(\theta) = 0\} = \{\theta \in \Theta : \langle g_k(\theta), u \rangle - \theta \leq 0 \ \forall u \in \Theta\}. \quad (16)$$

Because J_k is continuously differentiable on compact convex Θ , each J_k attains a maximizer satisfying (16); continuity of Q_k then makes S_k a nonempty compact set. At the realized iterate, let

$$\zeta_k := \|Q_k(\theta_k)\|_2^2, \quad \mathcal{E}_K := \frac{1}{K} \sum_{k=1}^K \mathbb{E} \zeta_k. \quad (17)$$

For window length $w \geq 1$ and discount $\rho \in (0, 1]$, set

$$W_{\rho, w} := \sum_{j=0}^{w-1} \rho^j, \quad \bar{Q}_k := W_{\rho, w}^{-1} \sum_{j=0}^{w-1} \rho^j Q_{k-j}(\theta_{k-j}), \quad (18)$$

with $Q_t(\theta_t) := 0$ for $t \leq 0$. The dynamic local regret (DLR) is

$$\mathcal{R}_{\rho, w}(K) := \sum_{k=1}^K \|\bar{Q}_k\|_2^2. \quad (19)$$

Here Q_k is the projected residual, ζ_k is the squared projected residual, and $\mathcal{E}_K$ is the average expected squared projected residual. Each projected residual is evaluated at its contemporaneous iterate before the residual vectors are exponentially averaged over the recent window. This is the dynamic local-regret object used to evaluate online adaptation below.

3.6 Assumptions

3.6.1 Standing statistical assumptions

Assumption 3.1 (Bounded analysis domain and likelihood regularity). *The parameter set $\Theta \subset \mathbb{R}^d$ is nonempty, compact, and convex, and is contained in an open set on which the policy is defined. The horizon h is finite and $N \geq 1$. There exist finite constants Q_f, B_X, B_r, B_v such that, on the analyzed domain,*

$$\sup_{i, u} \|q_{i, u}\|_2 \leq Q_f, \quad 0 < X_u \leq B_X, \quad 0 \leq r_u \leq B_r, \quad 0 \leq v_{\pm}(Y_k) \leq B_v. \quad (20)$$

Conditional on the observed history and action, the market transition kernel and the map $\mathbf{M}_u$ do not depend explicitly on θ . The latent Dirichlet proposals are retained in the augmented trajectory. Hence all explicit θ -dependence of the conditional trajectory law enters through the Dirichlet policy density in (6). The bounds and parameter-independence conditions above are required uniformly for every analyzed true episode model P_k and frozen simulator model $\tilde{P}_k$, conditional on the episode-start information and context. In particular, the frozen simulator is held fixed as θ varies.

Assumption 3.2 (Fixed regularized CPT). *The preference parameters in (9)–(10) are fixed. The functions $w_{\pm}^{\varepsilon}$ are strictly increasing and belong to $C^3([0, 1])$, with*

$$C_j := \max_{\sigma \in \{+, -\}} \sup_{p \in [0, 1]} |(w_{\sigma}^{\varepsilon})^{(j)}(p)| < \infty, \quad j = 1, 2, 3. \quad (21)$$

Under Assumptions 3.1–3.2, Appendix A derives finite uniform score moments and constants $B_g, L < \infty$ such that

$$\sup_{k, \theta \in \Theta} \|g_k(\theta)\|_2 \leq B_g, \quad \|g_k(\theta) - g_k(\theta')\|_2 \leq L\|\theta - \theta'\|_2. \quad (22)$$

3.6.2 Online variation and optional geometric conditions

Assumption 3.3 (Frozen-simulator error and objective variation). *For the queried parameter θ_k in episode k ,*

$$\|\tilde{g}_k(\theta_k) - g_k(\theta_k)\|_2 \leq \beta_k, \quad \mathbb{E}\beta_k^2 < \infty, \quad (23)$$

where β_k is $\mathcal{F}_k$ -measurable. Define

$$\mathcal{V}_K := \sum_{k=1}^{K-1} \sup_{\theta \in \Theta} |J_{k+1}(\theta, r_{k+1}) - J_k(\theta, r_k)|. \quad (24)$$

For the finite horizon under study, $\mathbb{E}\mathcal{V}_K < \infty$.

Assumption 3.4 (Stationary-set error bound; optional). *There exists $\kappa < \infty$ such that, uniformly over the analyzed episodes,*

$$\text{dist}(\theta, \mathcal{S}_k) \leq \kappa \|Q_k(\theta)\|_2, \quad \theta \in \Theta. \quad (25)$$

Assumption 3.5 (Reference-isolated coupling; optional). *When two episode-start references are compared, the market law, non-reference start state, and policy parameter are held fixed; the policy features and market transition kernel do not depend directly on the reference value. The two trajectories can therefore be coupled with common policy and market random variables, yielding the same wealth and portfolio paths.*

3.6.3 Fixed-target minimax experiment

Assumption 3.6 (Exact-trajectory minimax experiment). *Fix d , an interior point $\theta_0 = 0 \in \Theta$, finite upper bounds on h and N , and the common constants in Assumptions 3.1–3.2. Let $\mathfrak{C}$ be a class of conditional portfolio models satisfying these bounds and containing the Bernoulli-return submodel constructed in Appendix E; assume $(w_+^{\varepsilon})'(0) > 0$.*

An estimator may adaptively query any $\theta \in \Theta$. Each oracle call returns one fresh complete augmented trajectory from the queried policy under the unknown $P \in \mathfrak{C}$. The procedure may randomize and stop at a data-dependent call count T_{call} , subject to

$$\sup_{P \in \mathfrak{C}} \mathbb{E}_P T_{\text{call}} \leq B. \quad (26)$$

No additional observation of the unknown market law is available. The target is $g_P = \nabla J_P(\theta_0)$, while d and all domain constants remain fixed as $B \rightarrow \infty$.

Table 1: Role, verification route, and scope fallback of the assumptions. The fallback column states which conclusion is retained when the corresponding optional or asymptotic condition is unavailable.

Assumption	Role	Verification route	If unavailable
3.1	score identity, moments, smoothness	parameter box; bounded factors; Appendix A	re-establish the required likelihood moments on the enlarged domain
3.2	Taylor expansion and fixed-target debiasing	analytic derivatives of the fixed regularized weights	change the regularization or prove a replacement remainder bound
3.3	true-target online residual bound	simulator validation for β_k ; direct objective-difference bound for $\mathcal{V}_K$	finite-horizon theorem remains with the observed error terms; vanishing tracking is not asserted
3.4	residual $\Rightarrow$ distance to S_k	objective-specific geometry or a proved local error bound	retain residual and DLR conclusions only
3.5	explicit reference-only sensitivity decomposition	verify that features and market kernel do not use r directly	retain the general variation budget $\mathcal{V}_K$ without the reference-only split
3.6	fixed- d minimax information rate	check that the information source and budget match the oracle experiment	omit the minimax claim for richer-information settings

3.6.4 Assumption roles and scope

4 Algorithm

4.1 Centered leave-one-out base and raw randomized correction

Fix one frozen episode context and parameter, and suppress the episode index. A complete trajectory yields

$$\mathbf{X}_i = (V_i^+, V_i^-, G_i), \quad V_i^\pm = v_\pm(Y_i). \quad (27)$$

For $\sigma \in \{+, -\}$, let $\varsigma_+ = 1$, $\varsigma_- = -1$ and define

$$I_{i,\sigma}(y) = \mathbf{1}\{V_i^\sigma > y\}, \quad \hat{S}_{-i,n}^\sigma(y) = \frac{1}{n-1} \sum_{j \neq i} I_{j,\sigma}(y). \quad (28)$$

The centered leave-one-out statistic is

$$U_n^{\text{cen}} = \frac{1}{n} \sum_{i=1}^n G_i \sum_{\sigma \in \{+,-\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) \{I_{i,\sigma}(y) - \hat{S}_{-i,n}^\sigma(y)\} dy. \quad (29)$$

The corresponding raw statistic is

$$U_n^{\text{raw}} = \frac{1}{n} \sum_{i=1}^n G_i \sum_{\sigma \in \{+,-\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) I_{i,\sigma}(y) dy. \quad (30)$$

Choose $n \geq 2$, activation probability $\varrho \in (0, 1]$, and exponent $b \in (1, 2)$. Let

$$\nu_\ell = (1 - 2^{-b})2^{-b(\ell-1)}, \quad \ell = 1, 2, \dots \quad (31)$$

Algorithm 1 One centered-LOO debiased CPT-gradient output

Require: $(z, r, \tilde{P}, \theta)$, $n \geq 2$, $\varrho \in (0, 1]$, $b \in (1, 2)$

- 1: $\mathcal{X}_0 \leftarrow \{\mathbf{X}_i\}_{i=1}^n$ i.i.d. from the frozen law at (z, r, θ) ; $U_0 \leftarrow U_n^{\text{cen}}(\mathcal{X}_0)$
- 2: $\mathbf{A} \sim \text{Bernoulli}(\varrho)$
- 3: **if** $\mathbf{A} = 0$ **then**
- 4: **return** U_0
- 5: **end if**
- 6: $\ell \sim \nu$; $\mathcal{X}_\ell \leftarrow \{\mathbf{X}_i\}_{i=1}^{n2^\ell}$ i.i.d. from the same frozen law, independently of $\mathcal{X}_0$
- 7: $D \leftarrow U_{n2^\ell}^{\text{raw}}(\mathcal{X}_\ell) - \frac{1}{2}\{U_{n2^{\ell-1}}^{\text{raw}}(\mathcal{X}_\ell^{(1)}) + U_{n2^{\ell-1}}^{\text{raw}}(\mathcal{X}_\ell^{(2)})\}$
- 8: **return** $Z_n^{\text{db}} = U_0 + D/(\varrho\nu_\ell)$

For $n_\ell = n2^\ell$, compute $U_{n_\ell}^{\text{raw}}$ on one batch of n_ℓ trajectories and the two half-sample statistics on its two disjoint halves. Define

$$D_{n,\ell}^{\text{raw}} = U_{n_\ell}^{\text{raw}} - \frac{1}{2} \left(U_{n_{\ell-1}}^{\text{raw},(1)} + U_{n_{\ell-1}}^{\text{raw},(2)} \right). \quad (32)$$

Let $\mathbf{A} \sim \text{Bernoulli}(\varrho)$ and, conditional on $\mathbf{A} = 1$, let $\ell^* \sim \nu$; both are independent of the base sample and the correction batch. One complete debiased output is

$$Z_n^{\text{db}} = U_n^{\text{cen}} + \frac{\mathbf{A}}{\varrho\nu_{\ell^*}} D_{n,\ell^*}^{\text{raw}}, \quad (33)$$

with the second term defined as zero when $\mathbf{A} = 0$.

The level distribution is untruncated: no deterministic cap is imposed on ℓ^* , and completed outputs are not discarded as a function of the realized level or magnitude.

4.2 Online update

In episode k , generate M_k conditionally independent copies of (33) under the frozen simulator and set

$$\hat{g}_k^{\text{db}} = \frac{1}{M_k} \sum_{m=1}^{M_k} Z_{k,m}^{\text{db}}. \quad (34)$$

The parameter update is

$$\theta_{k+1} = \Pi_\Theta \left\{ \theta_k + \gamma \mathbf{d}(\hat{g}_k^{\text{db}}) \right\}. \quad (35)$$

5 Main Results

5.1 Reference dependence

Proposition 5.1 (Same terminal wealth, different CPT value). *Let $x_{\text{lo}} < r_0 < x_{\text{f}} < x_{\text{hi}}$ and consider*

$$\text{I} : r_0 \rightarrow x_{\text{hi}} \rightarrow x_{\text{f}}, \quad \text{II} : r_0 \rightarrow x_{\text{lo}} \rightarrow x_{\text{f}}.$$

Assume

$$x_{\text{f}} < (1 - \eta_+)r_0 + \eta_+x_{\text{hi}}, \quad x_{\text{f}} > (1 - \eta_-)r_0 + \eta_-x_{\text{lo}}.$$

Then

$$r_{\text{I}} - r_{\text{II}} = \eta_+(1 - \eta_-)x_{\text{hi}} - \eta_-(1 - \eta_+)x_{\text{lo}} + (\eta_- - \eta_+)x_{\text{f}}. \quad (36)$$

Algorithm 2 Dynamic-reference CPT policy gradient (DRCPT-PG)

Require: $\theta_1 \in \Theta$, (X_1, r_1, ω_0) , $h, \chi, \gamma, \vartheta$, (n, ϱ, b) , $\{M_k\}_{k=1}^K$

- 1: **for** $k = 1, \dots, K$ **do**
- 2: fix $(z_k, r_k, \tilde{P}_k)$ conditionally on $\mathcal{F}_k$
- 3: generate $Z_{k,1}^{\text{db}}, \dots, Z_{k,M_k}^{\text{db}}$ independently by Algorithm 1
- 4: $\hat{g}_k^{\text{db}} \leftarrow M_k^{-1} \sum_{m=1}^{M_k} Z_{k,m}^{\text{db}}$
- 5: **for** $u = t_k, \dots, t_k + h - 1$ **do**
- 6: $\tilde{\omega}_u \sim \pi_{\theta_k}(\cdot \mid s_u)$; $\omega_u \leftarrow (1 - \chi)\omega_{u-1} + \chi \mathbf{M}_u(\tilde{\omega}_u)$
- 7: $X_{u+1} \leftarrow X_u \{1 + \omega_u^\top R_{u+1} - c_{\text{tc}} \|\omega_u - \omega_{u-1}\|_1\}$; $r_{u+1} \leftarrow \Phi(r_u, X_{u+1})$
- 8: **end for**
- 9: $\theta_{k+1} \leftarrow \Pi_\Theta\{\theta_k + \gamma \mathbf{d}(\hat{g}_k^{\text{db}})\}$
- 10: **end for**
- 11: **return** $\{\theta_k, X_{t_k}, r_k, \omega_{t_k-1}\}_{k=1}^{K+1}$

If the right-hand side is nonzero, the two paths have different regularized CPT values despite identical terminal wealth x_{f} .

Proposition 5.2 (Reference sensitivity). *Under Assumptions 3.1, 3.2, and 3.5, set*

$$\eta_{\min} = \min(\eta_+, \eta_-), \quad L_v = \alpha_+ \varepsilon_v^{\alpha_+ - 1} + \lambda \alpha_- \varepsilon_v^{\alpha_- - 1}. \quad (37)$$

For two episode-start references r, r' coupled through the same wealth path,

$$|r_{t_k+h}(r) - r_{t_k+h}(r')| \leq (1 - \eta_{\min})^h |r - r'|, \quad (38)$$

and, for fixed (P, z, θ) ,

$$|J(P, z, r; \theta) - J(P, z, r'; \theta)| \leq C_1 L_v (1 - \eta_{\min})^h |r - r'|. \quad (39)$$

Consequently, with

$$\begin{aligned} v_k^{\text{nr}} &:= \sup_{\theta \in \Theta} |J(P_{k+1}, z_{k+1}, r_k; \theta) - J(P_k, z_k, r_k; \theta)|, \\ \sup_{\theta \in \Theta} |J_{k+1}(\theta, r_{k+1}) - J_k(\theta, r_k)| &\leq v_k^{\text{nr}} + C_1 L_v (1 - \eta_{\min})^h |r_{k+1} - r_k|. \end{aligned} \quad (40)$$

5.2 Gradient identity and debiased estimation

Theorem 5.3 (Regularized CPT score-gradient identity). *Under Assumptions 3.1–3.2, fix one target and write $J = J_k$, $Y = Y_k$, $S_\theta^\pm(y) := S_{P_k, z_k, r_k, \theta}^\pm(y)$, and $G_\theta := G_{k, \theta}$. Define*

$$\begin{aligned} \Psi_\theta(\tau) &= \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \{ \mathbf{1}[v_+(Y(\tau)) > y] - S_\theta^+(y) \} dy \\ &\quad - \int_0^{B_v} (w_-^\varepsilon)'(S_\theta^-(y)) \{ \mathbf{1}[v_-(Y(\tau)) > y] - S_\theta^-(y) \} dy. \end{aligned} \quad (41)$$

Then

$$\nabla J(\theta) = \mathbb{E}\{\Psi_\theta(\tau) G_\theta(\tau)\}. \quad (42)$$

The same identity holds for the frozen simulator objective $\tilde{J}$.

Lemma 5.4 (Coupled full/half cancellation). *Under Assumptions 3.1–3.2, fix one target and parameter and write $g := \nabla J(\theta)$. There exist a measurable $\mathfrak{h}$ and constants $C_{\text{rem}}, C_U < \infty$ such that*

$$U_n^{\text{raw}} - g = \frac{1}{n} \sum_{i=1}^n \mathfrak{h}(X_i) + \mathcal{B}_n, \quad \mathbb{E}\mathfrak{h}(X_i) = 0, \quad \mathbb{E}\|\mathcal{B}_n\|_2^2 \leq \frac{C_{\text{rem}}}{n^2}, \quad (43)$$

which implies

$$\mathbb{E}\|D_{n,\ell}^{\text{raw}}\|_2^2 \leq \frac{9C_{\text{rem}}}{n^2} 2^{-2\ell}, \quad \ell \geq 1, \quad (44)$$

and

$$\text{tr Cov}(U_n^{\text{cen}}) \leq \frac{C_U}{n}. \quad (45)$$

Lemma 5.5 (Centering and finite-sample bias). *For $n \geq 2$,*

$$\mathbb{E}U_n^{\text{cen}} = \mathbb{E}U_n^{\text{raw}} =: \mu_n, \quad (46)$$

and there exists $C_{\text{bias}} < \infty$ such that

$$\|\mu_n - g\|_2 \leq \frac{C_{\text{bias}}}{n}, \quad g := \nabla J(\theta). \quad (47)$$

Theorem 5.6 (Exact unbiasedness, finite variance, and finite expected cost). *Let Z_n^{db} be defined by (33), with $n \geq 2$, $\varrho \in (0, 1]$, and $1 < b < 2$. Put*

$$\Xi_b := \sum_{\ell \geq 1} \frac{4^{-\ell}}{\nu_\ell} = \frac{1}{4(1 - 2^{-b})(1 - 2^{b-2})}, \quad \Upsilon_b := \sum_{\ell \geq 1} \nu_\ell 2^\ell = \frac{2(1 - 2^{-b})}{1 - 2^{1-b}}. \quad (48)$$

Then

$$\mathbb{E}Z_n^{\text{db}} = g, \quad (49)$$

$$\text{tr Cov}(Z_n^{\text{db}}) \leq C_{\text{var}}(n) := \frac{C_U}{n} + \frac{9C_{\text{rem}}\Xi_b}{\varrho n^2} < \infty, \quad (50)$$

and the expected number of complete trajectories is

$$C_{\text{cost}}(n, \varrho, b) = n(1 + \varrho\Upsilon_b) < \infty. \quad (51)$$

For M independent outputs,

$$\hat{g}_{M,n}^{\text{db}} := \frac{1}{M} \sum_{m=1}^M Z_{n,m}^{\text{db}}, \quad \mathbb{E}\|\hat{g}_{M,n}^{\text{db}} - g\|_2^2 \leq \frac{C_{\text{var}}(n)}{M}. \quad (52)$$

Theorem 5.7 (Fixed-target asymptotic normality). *Under Theorem 5.6, fix (n, ϱ, b) and let $\Sigma_n = \text{Cov}(Z_n^{\text{db}})$. Then*

$$\sqrt{M}(\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, \Sigma_n). \quad (53)$$

If Σ_n is nonsingular and $\hat{\Sigma}_{M,n}$ is the empirical covariance of the M outputs, then

$$\sqrt{M}\hat{\Sigma}_{M,n}^{-1/2}(\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, I_d). \quad (54)$$

5.3 Online residual, dynamic local regret, and stationary-set tracking

Lemma 5.8 (One-step normalized projected ascent). *Let J be L -smooth on the nonempty closed convex set Θ . Fix $x \in \Theta$, let $g = \nabla J(x)$ with $\|g\|_2 \leq B_g$, and let $\hat{g}$ be any finite vector. Set*

$$x^+ = \Pi_\Theta\{x + \gamma d(\hat{g})\}, \quad Q(x) = \gamma^{-1}[\Pi_\Theta\{x + \gamma d(g)\} - x].$$

If $\gamma L \leq \vartheta$ and $C_g := \max\{B_g, \vartheta\}$, then

$$J(x^+) - J(x) \geq \frac{\vartheta\gamma}{8}\|Q(x)\|_2^2 - \frac{5C_g\gamma}{4\vartheta^2}\|\hat{g} - g\|_2^2. \quad (55)$$

Theorem 5.9 (Average expected squared projected residual bound). *Suppose Assumptions 3.1–3.3 hold and $\gamma L \leq \vartheta$. Let $\hat{g}_k^{\text{db}}$ be defined by (34) and target $\tilde{g}_k(\theta_k)$. Then*

$$\mathbb{E}\|\hat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2 \leq \frac{C_{\text{var}}(n)}{M_k} + \mathbb{E}\beta_k^2, \quad (56)$$

and

$$\mathcal{E}_K \leq \frac{8\{2B_v + \mathbb{E}\mathcal{V}_K\}}{\vartheta\gamma K} + \frac{10C_g}{\vartheta^3 K} \sum_{k=1}^K \left(\frac{C_{\text{var}}(n)}{M_k} + \mathbb{E}\beta_k^2 \right), \quad (57)$$

Corollary 5.10 (Dynamic local regret). *For every $K \geq 1$, pathwise,*

$$\mathcal{R}_{\rho,w}(K) \leq \sum_{k=1}^K \zeta_k, \quad (58)$$

and therefore

$$\frac{1}{K} \mathbb{E}\mathcal{R}_{\rho,w}(K) \leq \mathcal{E}_K. \quad (59)$$

Corollary 5.11 (Moving stationary-set tracking). *If Assumption 3.4 also holds, then*

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E} \text{dist}^2(\theta_k, \mathbf{S}_k) \leq \kappa^2 \mathcal{E}_K. \quad (60)$$

Lemma 5.12 (Drift of stationary sets). *Under Assumption 3.4, define*

$$\delta_k^g := \sup_{\theta \in \Theta} \|g_{k+1}(\theta) - g_k(\theta)\|_2. \quad (61)$$

Then

$$d_H(\mathbf{S}_{k+1}, \mathbf{S}_k) \leq \frac{\kappa}{\vartheta} \delta_k^g. \quad (62)$$

Corollary 5.13 (Vanishing average expected squared projected residual and DLR). *Fix $w < \infty$ and $\rho \in (0, 1]$. If*

$$\mathbb{E}\mathcal{V}_K = o(K), \quad \frac{1}{K} \sum_{k=1}^K \left(\frac{1}{M_k} + \mathbb{E}\beta_k^2 \right) \rightarrow 0, \quad (63)$$

then

$$\mathcal{E}_K \rightarrow 0, \quad K^{-1} \mathbb{E}\mathcal{R}_{\rho,w}(K) \rightarrow 0. \quad (64)$$

Under Assumption 3.4, the left-hand side of (60) also converges to zero.

The growing replication schedule used in E3 is

$$M_k = \max\{4, \lceil \sqrt{k} \rceil\}, \quad k \geq 1. \quad (65)$$

Since $M_k \geq \sqrt{k}$ and $\sum_{k=1}^K k^{-1/2} \leq 1 + \int_1^K x^{-1/2} dx = 2\sqrt{K} - 1$,

$$\frac{1}{K} \sum_{k=1}^K \frac{1}{M_k} \leq \frac{2\sqrt{K} - 1}{K} = O(K^{-1/2}). \quad (66)$$

Thus the sampling term in (57) vanishes under this schedule. Under the remaining hypotheses of [Theorem 5.9](#), if $\mathbb{E}\mathcal{V}_K = o(K)$ and $K^{-1} \sum_{k=1}^K \mathbb{E}\beta_k^2 \rightarrow 0$, [Corollary 5.13](#) yields sublinear expected dynamic local regret.

5.4 Fixed-dimension minimax information rate

Under Assumption [3.6](#), define

$$\mathfrak{R}_B := \inf_{\hat{g}: \sup_{P \in \mathfrak{C}} \mathbb{E}_P T_{\text{call}} \leq B} \sup_{P \in \mathfrak{C}} \mathbb{E}_P \|\hat{g} - g_P\|_2^2. \quad (67)$$

Theorem 5.14 (Minimax-optimal fixed-dimension trajectory-budget rate). *Under Assumption [3.6](#), there exist constants $c_{\text{lb}}, C_{\text{ub}}, B_0 \in (0, \infty)$, independent of B , such that for all $B \geq B_0$,*

$$\frac{c_{\text{lb}}}{B} \leq \mathfrak{R}_B \leq \frac{C_{\text{ub}}}{B}. \quad (68)$$

The upper bound is attained, up to constants, by averaging independent untruncated outputs Z_n^{db} under the same expected complete-trajectory budget. Hence, for fixed d and fixed regularization, the minimax gradient-estimation rate is $\Theta(B^{-1})$.

6 Experimental Design and Evidence

6.1 Experimental roadmap

The experiments follow the argument developed in the theory. E1 isolates the path-dependent reference mechanism and its decision consequence. E2 then defines the shared semi-synthetic market setup and evaluates gradient estimation at fixed targets within that setup. E3 releases the same setup as a continuous slow-variation online process and tests whether continued policy updates track the moving first-order target. E4 perturbs the reference dynamics, episode horizon, and optimizer scale around the frozen E3 configuration to determine whether the tracking advantage occupies a parameter region rather than a single operating point. [Figure 1](#) summarizes this evidence chain; E5–E7 are reserved for real-market performance, investor behavior, and heterogeneous-agent interaction.

Experimental evidence map

Mechanism / Statistical estimation / Online adaptation / Robustness / External validity

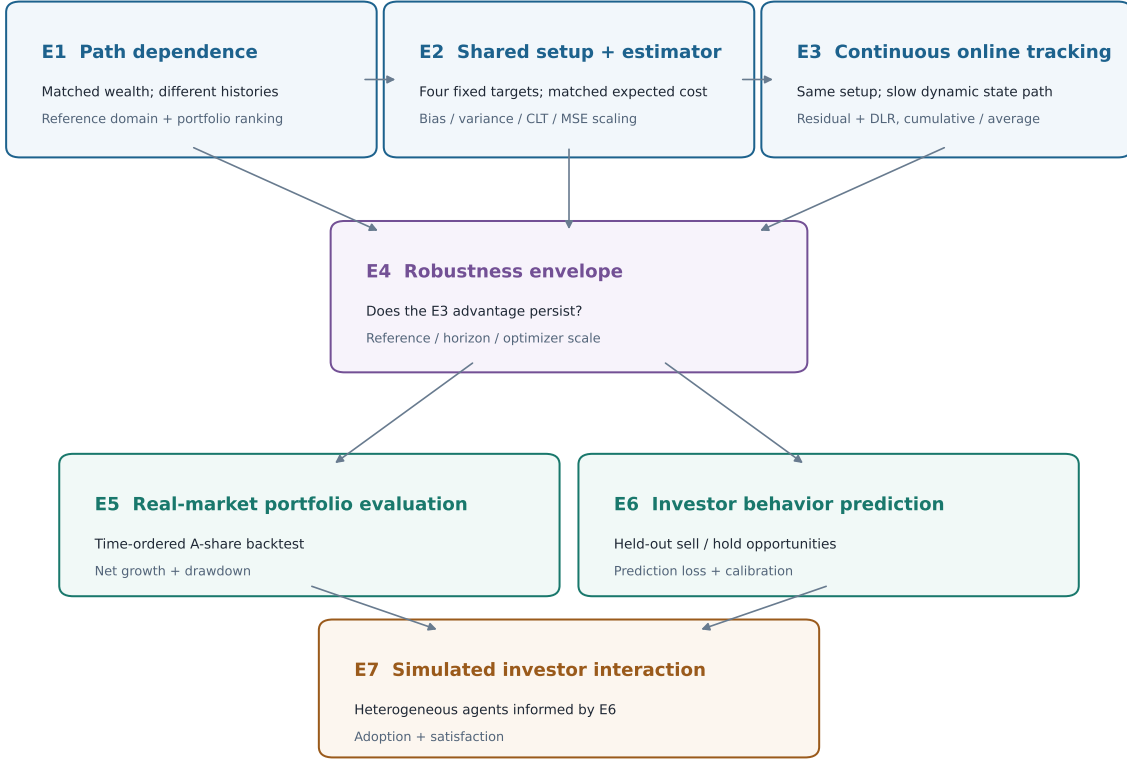


Figure 1: Experimental evidence map. E1 establishes the decision consequence of path-dependent reference formation. E2 evaluates the proposed gradient estimator at fixed targets in the shared semi-synthetic setup. E3 runs the same setup continuously under a theory-aligned slow-variation construction and measures first-order tracking by projected residual and dynamic local regret (DLR). E4 maps the robustness of the resulting online-tracking advantage.

6.2 E1: Path dependence and portfolio choice

Design. We construct two histories with the same initial wealth and the same endpoint,

$$(1, 1.10, 1.015) \quad \text{and} \quad (1, 0.90, 1.015),$$

and apply the reference recursion in (3) with $(\eta_+, \eta_-) = (0.20, 0.05)$. The histories are matched at the decision time: current wealth is $X_T = 1.015$, the pre-trade portfolio is 50% cash and 50% risky asset, and the next risky return is +4% with probability p and −4% otherwise. We compare two feasible target allocations, 20% and 80% risky. Under the action inertia in (7), these targets execute as 44% and 56% risky. Because the next-period distribution has two states, all CPT values are evaluated exactly rather than by Monte Carlo.

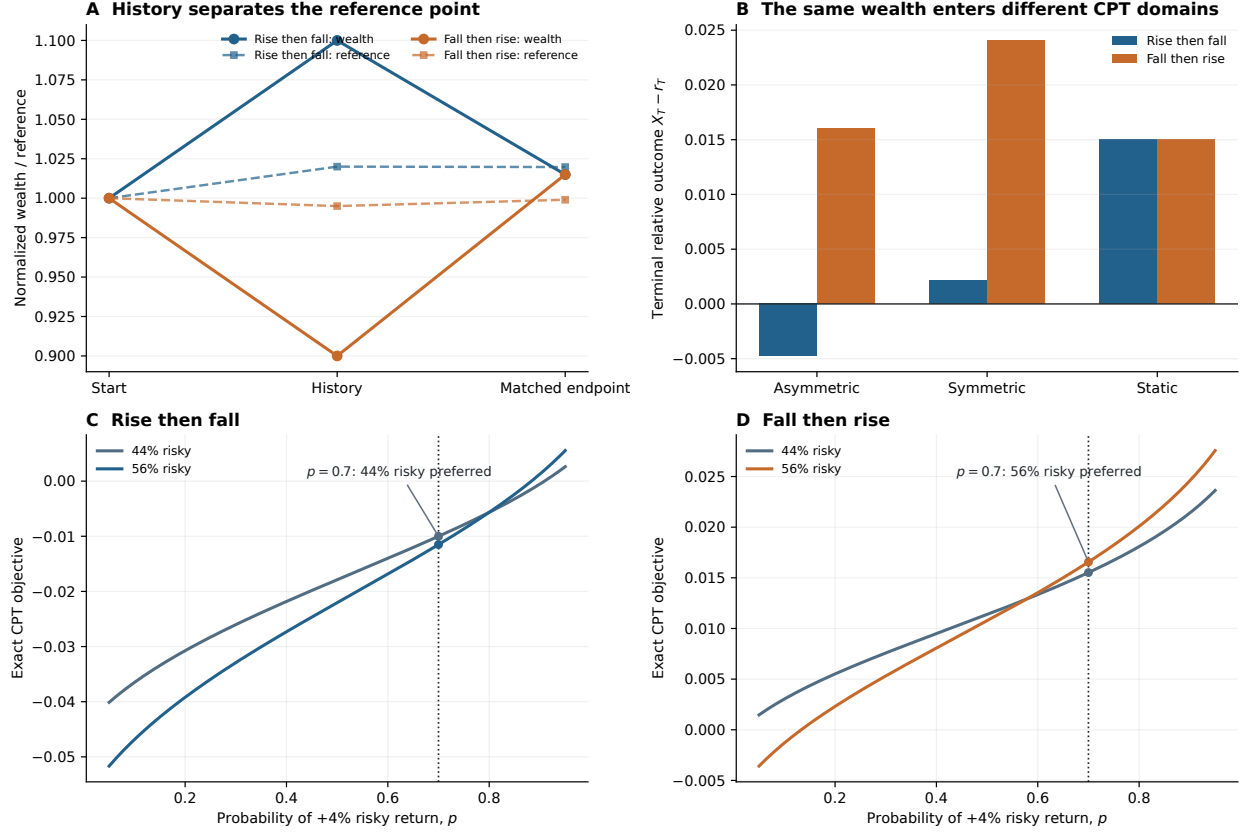


Figure 2: Path dependence changes both the perceived domain and the portfolio ranking. **A**: The two histories end at the same wealth but generate different asymmetric reference points. **B**: The asymmetric rule places the rise–fall history below its reference and the fall–rise history above its reference; the symmetric and static rules provide mechanism controls. **C–D**: Exact CPT values of the two feasible executed allocations as the probability p of a +4% risky return varies. At $p = 0.7$, the matched rise–fall account prefers 44% risky whereas the matched fall–rise account prefers 56% risky.

Table 2: E1 exact mechanism summary at $p = 0.7$.

History	X_T	r_T	$X_T - r_T$	Preferred risky share	CPT domain
Rise then fall	1.015	1.01975	−0.00475	44%	Loss
Fall then rise	1.015	0.99900	+0.01600	56%	Gain

Results. The same terminal wealth is evaluated on opposite sides of the CPT reference. After the rise–fall history, the reference is 1.01975, so $X_T - r_T = -0.00475$; after the fall–rise history it is 0.999, so $X_T - r_T = 0.016$. The separation propagates to the decision itself. At $p = 0.7$, the rise–fall history gives CPT values -0.01001 and -0.01151 for the 44% and 56% risky allocations, respectively, while the fall–rise history gives 0.01552 and 0.01656. Thus two investors with identical current wealth, holdings, and opportunity set rank the same feasible portfolios differently solely because their histories imply different reference points. This is the decision-level consequence of the path dependence characterized in [Proposition 5.1](#).

6.3 E2: Finite-sample gradient estimation

Shared semi-synthetic setup. Beginning with E2, we use a common 30-stock-plus-cash semi-synthetic market built from a frozen empirical A-share factor-block library. Factor blocks are sampled from a time-invariant empirical distribution, while the return law follows a pre-specified path across episodes. For risky asset i in episode k ,

$$R_{i,k,u} = 0.08 \tanh\left(\frac{\mu_k + 2.5 b_k^\top f_{i,u} + \xi_{k,u}^{\text{com}} + \xi_{i,k,u}^{\text{id}}}{0.08}\right),$$

where the common and idiosyncratic shocks have standard deviations $0.006s_k$ and $0.009s_k$, respectively. Episodes 1–40 use the stable momentum law, episodes 41–60 use the reversal law, episodes 61–120 interpolate to the terminal low-volatility/value/quality law, and the exogenous return law is fixed thereafter. E2 standardizes the account context to

$$(X, r, \omega, \theta) = (1, 1, e_{\text{cash}}, 0)$$

and freezes four pre-specified targets at episodes 20, 50, 90, and 150, corresponding to the stable, post-change, mid-drift, and post-drift phases. E3 subsequently uses the same factor distribution and episode-specific return law but allows the account state to evolve continuously.

We compare the proposed centered leave-one-out estimator with independent raw randomized correction against the raw independent-split plug-in estimator used by the adapted CPT-PG v5 baseline. Both target the square-smoothed CPT objective in (9). Expected complete-trajectory cost is matched over

$$B \in \{32, 64, 128, 256, 512, 1024, 2048\},$$

and numerical gradient references are generated independently at substantially higher cost.

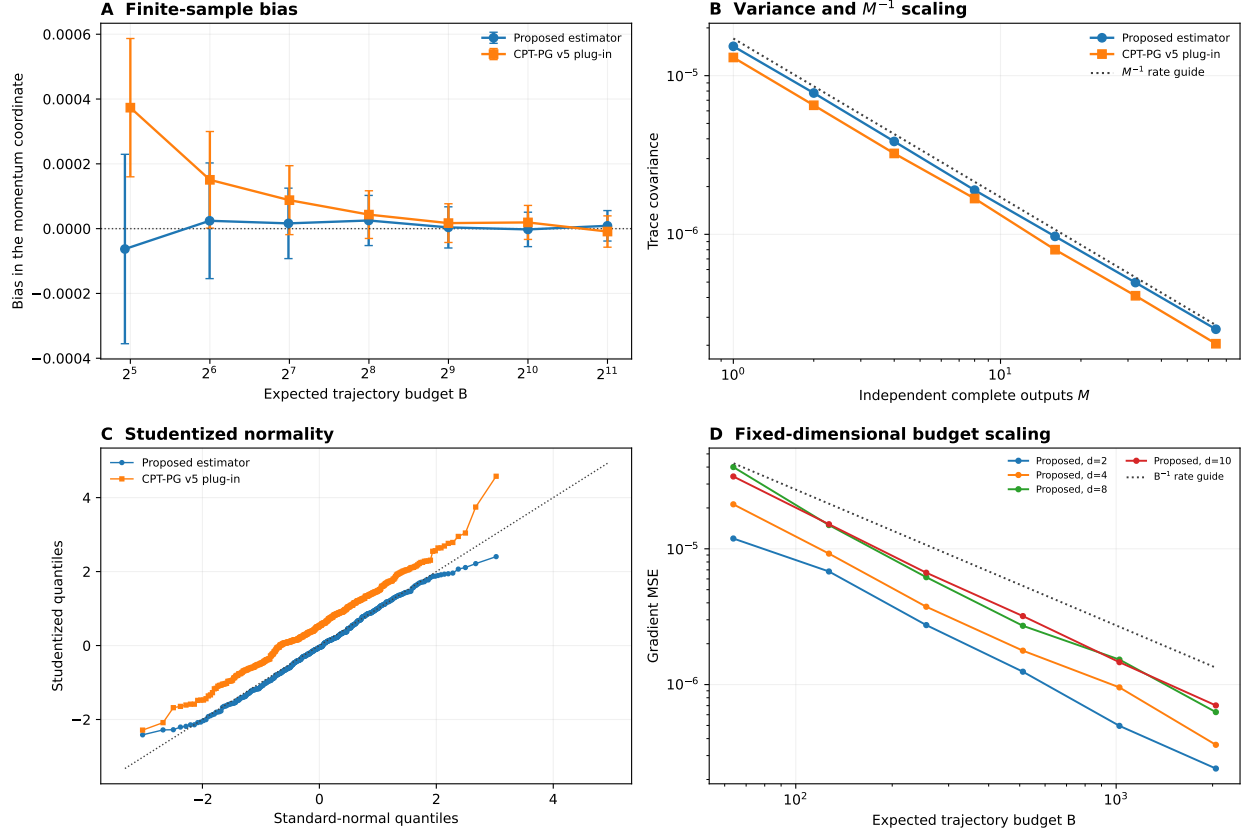


Figure 3: Finite-sample gradient estimation in the shared semi-synthetic setup. **A**: coordinate bias at the stable target. **B**: trace covariance of averages of M independent complete estimator outputs; the dotted reference has slope -1 . **C**: studentized normal-quantile diagnostic at $M = 64$. **D**: full-gradient MSE versus expected trajectory budget for $d \in \{2, 4, 8, 10\}$; the dotted reference has slope -1 . The corresponding unbiasedness, asymptotic normality, and fixed-dimensional minimax statements are given in [Theorems 5.6, 5.7](#) and [5.14](#).

Table 3: Full-gradient MSE at expected trajectory budget $B = 512$ across the four fixed targets. MSE entries are in units of 10^{-6} . The last column gives a paired 95% bootstrap interval for $\text{MSE}_{\text{prop}} - \text{MSE}_{\text{v5}}$ in the same units.

Target	Proposed	CPT-PG v5	Ratio	Difference 95% CI
Stable	3.123	2.938	1.063	$[-0.055, 0.434]$
Post-change	10.134	10.809	0.938	$[-1.495, 0.139]$
Mid-drift	10.647	11.468	0.928	$[-1.739, 0.102]$
Post-drift	10.589	12.257	0.864	$[-2.599, -0.721]$

Results. The four diagnostics separate estimator centering from ordinary Monte Carlo variance. At the stable target, the proposed coordinate estimate remains centered near the numerical reference across the budget range, while the plug-in estimator exhibits the larger low-budget displacement expected from nonlinear probability weighting (Figure 3A). Once independent complete outputs are averaged, both estimators recover the predicted inverse-replication variance rate: the fitted log-log slopes are -0.989 for the proposed estimator and -0.999 for CPT-PG v5 (Figure 3B). The proposed estimator also exhibits inverse-budget full-gradient MSE across $d = 2, 4, 8, 10$, with slopes between -1.17 and -1.12 , consistent with the fixed-dimensional $\Theta(B^{-1})$ rate in Theorem 5.14.

The studentized outputs sharpen the centering comparison. At $M = 64$, the proposed statistic has mean -0.049 , standard deviation 0.993 , and 95% coverage of 96.25% , whereas the v5 statistic has mean 0.556 and coverage of 91.75% (Figure 3C). At the standard budget $B = 512$, the proposed estimator has the lower MSE point estimate in three of the four fixed targets; the post-drift reduction is 13.6% and its paired interval lies entirely below zero (Table 3). Thus the debiased construction preserves the expected $1/B$ statistical rate while improving finite-sample centering and delivering competitive or lower full-gradient error across the pre-specified market states.

6.4 E3: Continuous slow-variation online tracking

Design. E3 releases the market setup introduced in E2 as one continuous online process. No account state is reset. If a five-day episode starting from (X_k, r_k, ω_k) produces the raw terminal state $(\tilde{X}_{k+1}, \tilde{r}_{k+1}, \tilde{\omega}_{k+1})$, the next episode starts from the diminishing-gain transition

$$\begin{aligned}\log X_{k+1} &= (1 - \alpha_k) \log X_k + \alpha_k \log \tilde{X}_{k+1}, \\ r_{k+1} &= (1 - \alpha_k) r_k + \alpha_k \tilde{r}_{k+1}, \\ \omega_{k+1} &= (1 - \alpha_k) \omega_k + \alpha_k \tilde{\omega}_{k+1}, \quad \alpha_k = k^{-3/4}.\end{aligned}$$

The within-episode reference update remains the asymmetric mechanism in (3) with $(\eta_+, \eta_-) = (0.20, 0.05)$. Hence the reference point retains its cross-episode memory, while bounded raw transitions imply episode-start state increments of order $O(\alpha_k)$ and

$$\sum_{k \leq K} \alpha_k = O(K^{1/4}) = o(K).$$

The factor-block distribution is time invariant, the exogenous return coefficients follow the E2 phase path through episode 120, and remain fixed for episodes 121–300. This construction keeps the dynamic-reference system continuous while imposing a sublinear state-step budget, providing the theory-aligned slow-variation setting used to probe Corollary 5.13.

The comparison uses 20 independent test trajectories. The online learner and frozen control share the complete path through episode 40. Thereafter the online learner continues Algorithm 2, whereas the control freezes θ but continues to propagate wealth, reference, and holdings under the same slow transition. We fix $(\gamma, \vartheta) = (0.08, 0.05)$ before the E3 test runs and use the growing replication schedule

$$M_k = \max\{4, \lceil \sqrt{k} \rceil\},$$

so that the average inverse replication count decays as $O(K^{-1/2})$ by (66). DLR follows (19) with $w = 5$ and $\rho = 0.9$. Two independent evaluation streams estimate both the squared projected residual and the squared norm of the window-averaged projected residual by cross-stream products, preventing the ordinary self-square noise term from entering the evaluation.

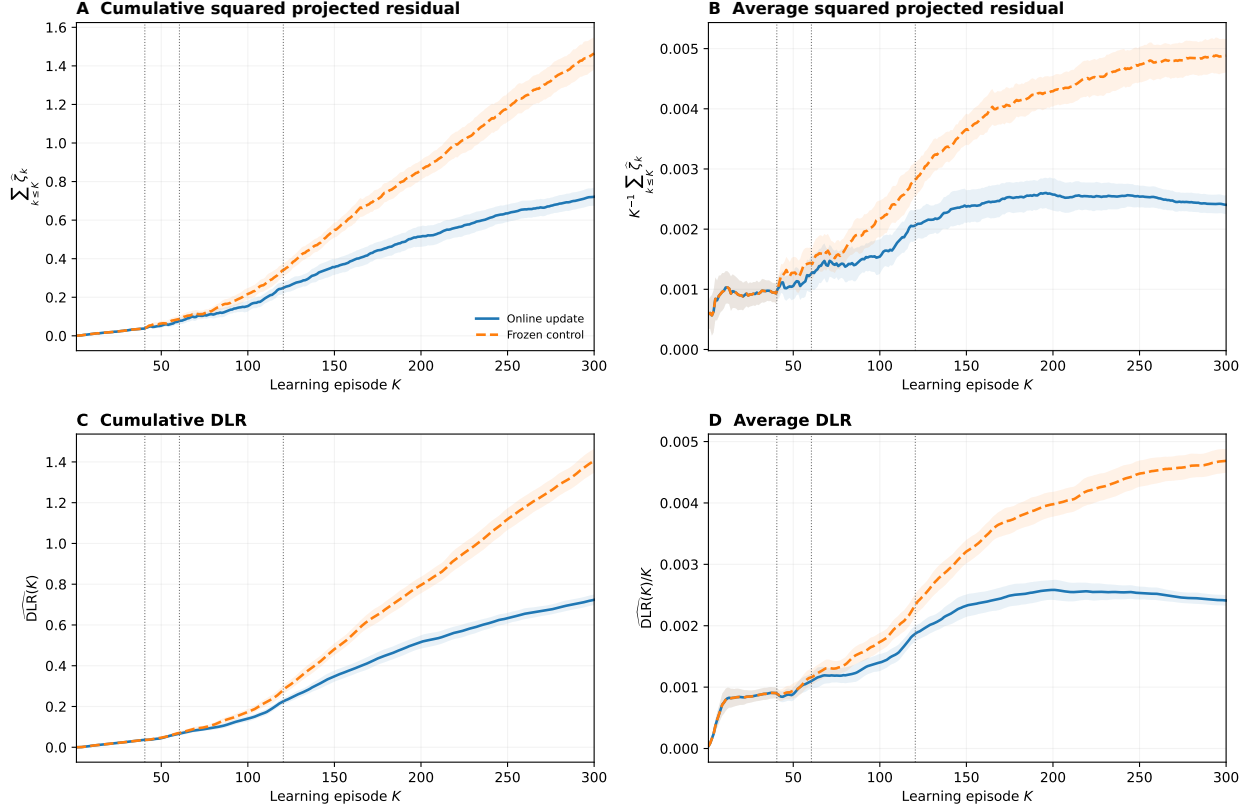


Figure 4: Continuous online tracking under slow variation. **A–B**: cumulative squared projected residual and its average. **C–D**: cumulative and average DLR. The two methods share the same trajectory through episode 40; thereafter the frozen control holds θ fixed while both methods continue to propagate their own wealth, reference, and holdings. Dotted lines mark the abrupt change, the start of gradual drift, and the point after which the exogenous return law is fixed. Curves are means over 20 independent trajectories and shaded bands are 1.96 standard errors.

Results. Continued updating produces substantially lower closed-loop tracking accumulation than freezing the policy parameter after episode 40. Over episodes 41–300, cumulative squared projected residual is 0.684 for the online learner and 1.427 for the frozen control, while cumulative DLR is 0.687 versus 1.369; the seed-paired differences have 95% bootstrap intervals $[-0.834, -0.650]$ and $[-0.738, -0.622]$, respectively. By episode 300, average DLR is 2.411×10^{-3} online and 4.685×10^{-3} frozen (Figure 4D). Because the policies induce different endogenous account paths after they diverge, the frozen curve is a closed-loop control rather than a common-objective optimizer comparison. The direct theory-aligned diagnostic is therefore the evolution of the squared projected residual and DLR along its own moving objective sequence.

The slow-variation tail also changes the growth rate of accumulated tracking error. Over episodes 121–300, the empirical log–log slope of cumulative DLR is 0.704 for continued updating and 0.968 for the frozen control; the corresponding cumulative squared projected residual slopes are 0.745 and 0.959. The online exponent below one is consistent with the sublinear direction predicted by Corollaries 5.10 and 5.13 for a slowly varying target sequence. Under the optional error-bound condition in Assumption 3.4, the average expected squared projected residual also controls the average squared distance from its policy iterates to the contemporaneous stationary set through Corollary 5.11.

6.5 E4: Robustness of online tracking

Design. E4 tests whether the E3 closed-loop tracking advantage persists when the reference dynamics, decision horizon, and optimizer scale are perturbed. Because each perturbation changes the induced objective sequence, absolute DLR values are not compared across configurations. Instead, within each configuration c we report the matched-control ratio

$$\mathcal{R}_{\text{DLR}}(c) = \frac{\text{DLR}_{\text{online}}^{\text{post}}(c)}{\text{DLR}_{\text{frozen}}^{\text{post}}(c)},$$

where the post-change interval begins at the first regime change. This is a within-configuration normalization of the same DLR used in E3, not a new loss; $\mathcal{R}_{\text{DLR}} < 1$ means that continued updating accumulates less DLR than freezing θ under the same configuration.

Panel A evaluates the full 9×9 reference grid

$$\eta_+, \eta_- \in \{0, 0.05, 0.10, \dots, 0.40\}.$$

Panel B varies the episode horizon over

$$h \in \{1, 2, 4, 5, 10, 20, 25, 50\},$$

while holding total market time at 1500 days and tying the state-gain clock to elapsed market time. Panel C evaluates 48 admissible optimizer configurations from

$$\gamma \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08, 0.10, 0.12\},$$

$$\vartheta \in \{0.005, 0.01, 0.015, 0.02, 0.03, 0.04, 0.05, 0.06\}, \quad \gamma L_{\text{diag}} \leq \vartheta, \quad L_{\text{diag}} = 0.31.$$

The E3 defaults $(\eta_+, \eta_-) = (0.20, 0.05)$, $h = 5$, and $(\gamma, \vartheta) = (0.08, 0.05)$ are marked but are not re-selected in E4. Each configuration uses four independent paired trajectories.

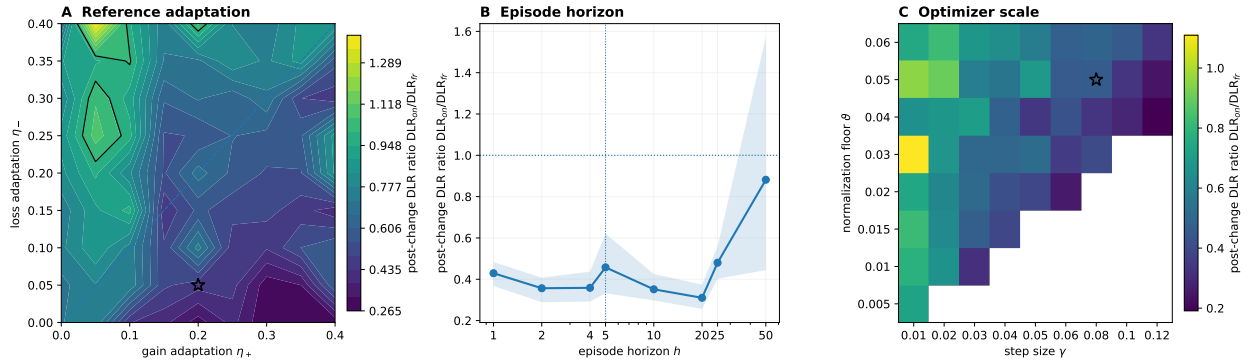


Figure 5: Robustness of the online-tracking advantage measured by the configuration-matched post-change DLR ratio $\mathcal{R}_{\text{DLR}}$. **A**: reference-adaptation surface over the simulated 9×9 grid; the black contour marks $\mathcal{R}_{\text{DLR}} = 1$, the diagonal marks $\eta_+ = \eta_-$, and the star marks the default asymmetric rule. **B**: horizon sweep with paired-seed 95% bootstrap intervals; the horizontal line marks $\mathcal{R}_{\text{DLR}} = 1$ and the vertical line marks $h = 5$. **C**: admissible (γ, ϑ) grid; blank cells are excluded by $\gamma L_{\text{diag}} > \vartheta$, and the star marks the E3 default.

Results. The reference-adaptation surface places most of the simulated grid below the matched-control boundary: 76 of 81 configurations have $\mathcal{R}_{\text{DLR}} < 1$, and 59 have paired 95% intervals entirely below one. All 36 asymmetric configurations with $\eta_+ > \eta_-$ lie below one, including 33 with intervals entirely below one. At the default rule (0.20, 0.05), the mean ratio is 0.458 with a 95% interval [0.332, 0.621]. The only region that clearly reverses the comparison is strongly loss-dominant adaptation; for example, (0.05, 0.40) gives a ratio of 1.403 with interval [1.224, 1.552] (Figure 5A).

The horizon sweep gives ratios below one at all eight tested horizons. Seven intervals lie entirely below one; the ratio ranges from 0.310 at $h = 20$ to 0.882 at $h = 50$, where the interval widens to [0.444, 1.572] and includes the neutral boundary (Figure 5B). The optimizer sweep shows the same broad pattern: 47 of 48 admissible configurations have mean ratios below one and 45 have intervals entirely below one. The single point estimate above one, $(\gamma, \vartheta) = (0.01, 0.03)$, has interval [0.876, 1.342] and is not resolved from the matched control. Together, the dense sweeps show that the E3 closed-loop tracking advantage extends over broad reference, horizon, and optimizer regions rather than depending on the default operating point.

6.6 E5: Real-market portfolio evaluation

[Placeholder: Time-ordered A-share backtest with return, risk, turnover, and cost metrics.]

6.7 E6: Investor behavior prediction

[Placeholder: Held-out investor sell/hold prediction and calibration.]

6.8 E7: Simulated heterogeneous-investor interaction

[Placeholder: Repeated interaction with heterogeneous simulated investor agents.]

7 Discussion

[Placeholder: Discussion.]

References

- [1] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979. doi: 10.2307/1914185.
- [2] Amos Tversky and Daniel Kahneman. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4):297–323, 1992. doi: 10.1007/BF00122574.
- [3] L. A. Prashanth, Cheng Jie, Michael C. Fu, Steven I. Marcus, and Csaba Szepesvári. Cumulative prospect theory meets reinforcement learning: Prediction and control. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1406–1415. PMLR, 2016.
- [4] Cheng Jie, L. A. Prashanth, Michael C. Fu, Steven I. Marcus, and Csaba Szepesvári. Stochastic optimization in a cumulative prospect theory framework. *IEEE Transactions on Automatic Control*, 63(9):2867–2882, 2018. doi: 10.1109/TAC.2017.2777469.

- [5] Olivier Lepel and Anas Barakat. Policy gradients for cumulative prospect theory in reinforcement learning. *Transactions on Machine Learning Research*, 2026. URL <https://arxiv.org/abs/2410.02605v5>. arXiv:2410.02605v5, version 5, revised 6 September 2026.
- [6] Hal R. Arkes, David Hirshleifer, Danling Jiang, and Sonya S. Lim. Reference point adaptation: Tests in the domain of security trading. *Organizational Behavior and Human Decision Processes*, 105(1):67–81, 2008. doi: 10.1016/j.obhdp.2007.04.005.
- [7] Hal R. Arkes, David Hirshleifer, Danling Jiang, and Sonya S. Lim. A cross-cultural study of reference point adaptation: Evidence from china, korea, and the US. *Organizational Behavior and Human Decision Processes*, 112(2):99–111, 2010. doi: 10.1016/j.obhdp.2010.02.002.
- [8] Xue Dong He and Linan Yang. Realization utility with adaptive reference points. *Mathematical Finance*, 29(2):409–447, 2019. doi: 10.1111/mafi.12182.
- [9] Cong Qin, L. H. Kong, and Xingye Yue. Realization utility with path-dependent reference points. *SIAM Journal on Financial Mathematics*, 13(3):912–948, 2022. doi: 10.1137/21M1411457.
- [10] Jianjun Gao, Yaoming Li, Yun Shi, and Jinyan Xie. Multi-period portfolio choice under loss aversion with dynamic reference point in serially correlated market. *Omega*, 127:103103, 2024. doi: 10.1016/j.omega.2024.103103.
- [11] Li An, Joseph Engelberg, Matthew Henriksson, Baolian Wang, and Jared Williams. The portfolio-driven disposition effect. *The Journal of Finance*, 79(5):3459–3495, 2024. doi: 10.1111/jofi.13378.
- [12] Siliu Chen and Fei Ren. Does social media information affect individual investor disposition effect? evidence from Xueqiu. *PLOS ONE*, 20(7):e0328547, 2025. doi: 10.1371/journal.pone.0328547.
- [13] Xiangyu Cui, Rujun Jiang, Yun Shi, Rufeng Xiao, and Yifan Yan. Decision making under cumulative prospect theory: An alternating direction method of multipliers. *INFORMS Journal on Computing*, 37(4):856–873, 2025. doi: 10.1287/ijoc.2023.0243.
- [14] Eric Luxenberg, Philipp Schiele, and Stephen Boyd. Portfolio optimization with cumulative prospect theory utility via convex optimization. *Computational Economics*, 64(5):3027–3047, 2024. doi: 10.1007/s10614-024-10556-x.
- [15] Zhongming Wu, Liu Yang, and Valentina De Simone. Behavioral portfolio optimization via cumulative prospect theory with a symmetric alternating direction method of multipliers. *Computational Optimization and Applications*, 93:303–334, 2026. doi: 10.1007/s10589-025-00721-9.
- [16] Zhebin Fan and Suman Neupane. Investor horizon, experience, and the disposition effect. *Journal of Behavioral and Experimental Finance*, 44:101003, 2024. doi: 10.1016/j.jbef.2024.101003.
- [17] Elad Hazan, Karan Singh, and Cyril Zhang. Efficient regret minimization in non-convex games. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1433–1441. PMLR, 2017.
- [18] Sergul Aydore, Tianhao Zhu, and Dean Foster. Dynamic local regret for non-convex online forecasting. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

- [19] Nadav Hallak, Panayotis Mertikopoulos, and Volkan Cevher. Regret minimization in stochastic non-convex learning via a proximal-gradient approach. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4008–4017. PMLR, 2021.
- [20] Songtao Feng, Ming Yin, Ruiquan Huang, Yu-Xiang Wang, Jing Yang, and Yingbin Liang. Non-stationary reinforcement learning under general function approximation. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 9976–10007. PMLR, 2023.
- [21] Guanghui Lan, Tianjiao Li, and Yangyang Xu. Projected gradient methods for nonconvex and stochastic smooth optimization: New complexities and auto-conditioned stepsizes, 2026. URL <https://arxiv.org/abs/2412.14291>. arXiv v2, revised 14 May 2026.
- [22] Chang-han Rhee and Peter W. Glynn. Unbiased estimation with square root convergence for SDE models. *Operations Research*, 63(5):1026–1043, 2015. doi: 10.1287/opre.2015.1404.
- [23] Pascal Massart. The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality. *The Annals of Probability*, 18(3):1269–1283, 1990. doi: 10.1214/aop/1176990746.
- [24] Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, 1998.
- [25] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.
- [26] Eric André and Guillaume Coqueret. Dirichlet policies for reinforced factor portfolios, 2020. URL <https://arxiv.org/abs/2011.05381>.
- [27] Xiao-Yang Liu, Ziyi Xia, Hongyang Yang, Jiechao Gao, Daochen Zha, Ming Zhu, Christina Dan Wang, Zhaoran Wang, and Jian Guo. Dynamic datasets and market environments for financial reinforcement learning. *Machine Learning*, 113:2795–2839, 2024. doi: 10.1007/s10994-023-06511-w.
- [28] Matthias X. Hanauer, Maarten Jansen, Laurens Swinkels, and Weili Zhou. Factor models for chinese A-shares. *International Review of Financial Analysis*, 91:102975, 2024. doi: 10.1016/j.irfa.2023.102975.
- [29] Kawin Ethayarajh, Winnie Xu, Niklas Muennighoff, Dan Jurafsky, and Douwe Kiela. Model alignment as prospect theoretic optimization. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 12634–12651. PMLR, 2024.
- [30] Jiaxin Liu, Yixuan Tang, Yi Yang, and Kar Yan Tam. Evaluating and aligning human economic risk preferences in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 18174–18188. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.917.
- [31] Sheyan Lalmohammed, Khush Gupta, Alok Shah, and Keshav Ramji. Modeling human behavior without humans: Prospect theoretic multi-agent reinforcement learning. In *ICML 2025 Workshop on Multi-Agent Systems in the Era of Foundation Models*, 2025. URL <https://openreview.net/forum?id=XDZdrTMWbM>.

- [32] Takehiro Takayanagi, Kiyoshi Izumi, Javier Sanz-Cruzado, Richard McCreadie, and Iadh Ounis. Are generative AI agents effective personalized financial advisors? In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 286–295. ACM, 2025. doi: 10.1145/3726302.3729897.
- [33] Balraj Verma, Mike Schulze, Divya Goswami, and Kamal Upreti. Artificial intelligence attitudes and resistance to use robo-advisors: Exploring investor reluctance toward cognitive financial systems. *Frontiers in Artificial Intelligence*, 8:1623534, 2025. doi: 10.3389/frai.2025.1623534.

A Notation and Analytic Preliminaries

All fixed-target statistical statements condition on the episode-start information, the queried parameter, and the frozen simulator before new Monte Carlo trajectories are drawn. Conditional independence is always with respect to this law. Limits in M fix the target and estimator design (n, ϱ, b) ; limits in K require the constants in the stated assumptions to be uniform over episodes; the minimax limit fixes d and all model-class constants while $B \rightarrow \infty$.

A.1 Main notation

Table 4: Main notation along the state, objective, estimation, and online-evaluation chain. Estimator internals and proof-local constants are defined where they are used.

Symbol	Type	Meaning
k, u, h	indices / integer	episode, calendar time, and finite episode horizon
$s_u, \omega_u, \tilde{\omega}_u$	state / portfolios	state $(m_u, X_u, r_u, \omega_{u-1})$ with market state m_u ; executed portfolio and latent proposal
$z_k, \mathcal{F}_k, r_k$	context / information	frozen non-reference start state, episode-start information, and reference r_{t_k}
$\theta, \Theta, \pi_\theta$	vector / set / policy	parameter, compact convex parameter set, and Dirichlet actor
$P_k, \tilde{P}_k$	episode models	true and frozen-simulator models inducing policy-dependent trajectory laws
$Y_k, v_\pm, w_\pm^\varepsilon$	outcome / functions	terminal relative outcome and regularized CPT value and weight maps
$J_k, \tilde{J}_k$	scalar functions	true and frozen-simulator CPT objectives
$g_k, \tilde{g}_k, G_{k,\theta}$	$\mathbb{R}^d$	true gradient, simulator gradient, and complete-trajectory score
$\hat{g}_k^{\text{db}}, M_k$	vector / integer	debaised update estimate averaged over M_k independent complete outputs
$Q_k; \gamma, \vartheta$	vector / scalars	projected residual; step size and normalization floor
$\mathcal{S}_k$	subset of Θ	constrained first-order stationary set
ζ_k	$\mathbb{R}_+$	squared projected residual $\ Q_k(\theta_k)\ _2^2$
$\mathcal{E}_K$	$\mathbb{R}_+$	average expected squared projected residual
$\mathcal{R}_{\rho,w}(K); \rho, w$	scalar / parameters	dynamic local regret (DLR); discount and window length
$\beta_k, \mathcal{V}_K$	$\mathbb{R}_+$	simulator-gradient error bound and cumulative true-objective variation
T_{call}, B	integer / scalar	oracle trajectory count and expected trajectory budget

A.2 Dirichlet score moments

Lemma A.1 (Uniform moments of the augmented trajectory score). *Under Assumption 3.1, there exist $0 < \underline{a} \leq \bar{a} < \infty$ such that*

$$\underline{a} \leq a_{i,u}(\theta) \leq \bar{a} \quad (69)$$

uniformly in i, u, θ . For every fixed $p < \infty$,

$$\sup_{k,\theta} \mathbb{E} \|G_{k,\theta}\|_2^p < \infty. \quad (70)$$

The trajectory log-likelihood Hessian $\mathcal{H}_{k,\theta}$ also satisfies

$$\sup_{k,\theta} \mathbb{E} \|\mathcal{H}_{k,\theta}\|_{\text{op}} < \infty. \quad (71)$$

Moreover, for every bounded measurable trajectory functional F , differentiation under the conditional trajectory law is valid up to order two:

$$\nabla_{\theta} \mathbb{E}_{\theta} F = \mathbb{E}_{\theta}(F G_{k,\theta}), \quad \nabla_{\theta}^2 \mathbb{E}_{\theta} F = \mathbb{E}_{\theta}\{F(G_{k,\theta} G_{k,\theta}^{\top} + \mathcal{H}_{k,\theta})\}. \quad (72)$$

Proof. Compactness of Θ and boundedness of $q_{i,u}$ imply (69). Every Dirichlet coordinate has a Beta marginal with parameters in a compact subset of $(0, \infty)^2$. Hence, for every finite p ,

$$\sup_{a,b} \int_0^1 |\log x|^p x^{a-1} (1-x)^{b-1} dx < \infty,$$

where the supremum is over the induced compact parameter set. The digamma and polygamma terms are bounded on the same positive compact interval. Equation (8), bounded features, and finite (h, N) therefore give (70); differentiating the score once more gives (71). The market kernel and $\mathbf{M}_u$ contribute no likelihood derivative by Assumption 3.1.

To justify differentiation under the trajectory integral, enlarge Θ slightly while keeping it compactly contained in the policy domain. On this neighborhood the Dirichlet concentrations remain in $[a_-, a_+]$ for some $a_- > 0$. For one latent action $\tilde{\omega}$, the Dirichlet density and its first two θ -derivatives are bounded, up to constants independent of θ , by

$$\prod_{i=0}^N \tilde{\omega}_i^{a_- - 1} \left(1 + \sum_{i=0}^N |\log \tilde{\omega}_i|^2 \right).$$

This envelope is integrable on the simplex for $a_- > 0$. Taking the finite product over the h latent actions gives an integrable trajectory envelope; the parameter-independent market kernels integrate to one conditionally at each step. Hence bounded F times either derivative is dominated by a common integrable function. Dominated convergence yields (72). $\square$

A.3 Smoothness of the CPT objective

Lemma A.2 (Uniform gradient and smoothness bounds). *Under Assumptions 3.1–3.2, there exist $B_g, L < \infty$ such that*

$$\sup_{k,\theta \in \Theta} \|g_k(\theta)\|_2 \leq B_g, \quad \|g_k(\theta) - g_k(\theta')\|_2 \leq L \|\theta - \theta'\|_2. \quad (73)$$

With

$$C_{G,2} := \sup_{k,\theta} \mathbb{E} \|G_{k,\theta}\|_2^2, \quad C_H := \sup_{k,\theta} \mathbb{E} \|\mathcal{H}_{k,\theta}\|_{\text{op}},$$

one admissible choice is

$$B_g = 2B_v C_1 \sqrt{C_{G,2}}, \quad L = 2B_v \{C_2 C_{G,2} + C_1 (C_{G,2} + C_H)\}. \quad (74)$$

Moreover $|J_k(\theta, r_k)| \leq B_v$.

Proof. For one side σ and threshold y , the likelihood-score identity and Lemma A.1 give

$$\nabla S_\theta^\sigma(y) = \mathbb{E}\{I_\sigma(y)G_\theta\}, \quad \nabla^2 S_\theta^\sigma(y) = \mathbb{E}\{I_\sigma(y)(G_\theta G_\theta^\top + \mathcal{H}_\theta)\}.$$

The bounded indicator and the moment bounds provide an integrable dominating function, so the derivative–expectation interchange is justified by dominated convergence. Consequently

$$\|\nabla S_\theta^\sigma(y)\|_2 \leq \sqrt{C_{G,2}}, \quad \|\nabla^2 S_\theta^\sigma(y)\|_{\text{op}} \leq C_{G,2} + C_H.$$

Differentiating the CPT integral twice and using Assumption 3.2 gives

$$\nabla^2 \int_0^{B_v} w_\sigma^\varepsilon(S_\theta^\sigma(y)) dy = \int_0^{B_v} \{(w_\sigma^\varepsilon)''(S) \nabla S \nabla S^\top + (w_\sigma^\varepsilon)'(S) \nabla^2 S\} dy.$$

The integrand is uniformly dominated by the constants above, giving (74). The first-derivative bound is analogous. Finally $0 \leq w_\sigma^\varepsilon \leq 1$ and the two integrals have common length B_v , so $|J_k| \leq B_v$. $\square$

A.4 Projected residual and constrained stationarity

Lemma A.3 (Projected residual characterizes first-order stationarity). *For closed convex Θ , $\gamma, \vartheta > 0$, and $g \in \mathbb{R}^d$,*

$$\Pi_\Theta\{\theta + \gamma d(g)\} = \theta \iff \langle g \rangle u - \theta \leq 0 \quad \forall u \in \Theta. \quad (75)$$

Hence the two definitions in (16) coincide.

Proof. The projection variational inequality gives $x = \Pi_\Theta(y)$ iff $\langle y - x \rangle u - x \leq 0$ for every $u \in \Theta$. Substitute $y = \theta + \gamma d(g)$ and use $d(g) = g / \max\{\|g\|_2, \vartheta\}$, a positive scalar multiple of g . $\square$

B Proofs for the Reference Mechanism and CPT Gradient

B.1 Same terminal wealth and path-dependent reference

Proof. It is enough to compute the two terminal references and then use strict monotonicity of the deterministic CPT value.

For path I, the first step is a gain, so its interim reference is

$$r_I^{(1)} = (1 - \eta_+)r_0 + \eta_+x_{\text{hi}}.$$

The first displayed condition in the proposition makes the second step a loss relative to this interim reference. Hence

$$r_I = (1 - \eta_-)\{(1 - \eta_+)r_0 + \eta_+x_{\text{hi}}\} + \eta_-x_f.$$

For path II, the first step is a loss and

$$r_{II}^{(1)} = (1 - \eta_-)r_0 + \eta_-x_{\text{lo}}.$$

The second displayed condition makes the move to x_f a gain, so

$$r_{II} = (1 - \eta_+)\{(1 - \eta_-)r_0 + \eta_-x_{\text{lo}}\} + \eta_+x_f.$$

Subtracting cancels the coefficient of r_0 and gives (36).

It remains to translate a nonzero reference gap into a CPT-value gap. A deterministic relative outcome y_0 has survival function equal to one below its corresponding value magnitude and zero above it. Since $w_\pm^\varepsilon(0) = 0$ and $w_\pm^\varepsilon(1) = 1$, its CPT value is $v_+(y) - v_-(y)$. The shifted-power gain term is strictly increasing on $y > 0$, and $-v_-(y)$ is strictly increasing on $y < 0$; the two pieces meet continuously at zero. Thus the signed deterministic CPT value is strictly increasing in y . If $r_I \neq r_{II}$, then $x_f - r_I \neq x_f - r_{II}$, so the two CPT values differ. $\square$

B.2 Reference contraction and CPT sensitivity

Proof. To prove (39), it is enough to establish three bounds: contraction of the reference recursion, Lipschitz continuity of the value transformation, and an L^1 survival-function coupling inequality.

Fix a wealth value x . The map $r \mapsto \Phi(r, x)$ is continuous, with slope $1 - \eta_+$ when $r \leq x$ and slope $1 - \eta_-$ when $r > x$. Splitting an interval at x when necessary therefore gives

$$|\Phi(r, x) - \Phi(r', x)| \leq (1 - \eta_{\min})|r - r'|.$$

Under Assumption 3.5, the two compared trajectories have the same wealth path. Iterating the preceding inequality for h steps proves (38). Since their terminal wealth is common,

$$|Y(r) - Y(r')| \leq (1 - \eta_{\min})^h |r - r'|. \quad (76)$$

For $x \geq 0$,

$$\frac{d}{dx} u_{\alpha, \varepsilon_v}(x) = \alpha x(x^2 + \varepsilon_v^2)^{\alpha/2-1} \leq \alpha \varepsilon_v^{\alpha-1},$$

where the bound uses $x \leq (x^2 + \varepsilon_v^2)^{1/2}$ and $\alpha \leq 1$. Hence

$$|v_+(y) - v_+(y')| + |v_-(y) - v_-(y')| \leq L_v |y - y'|.$$

Combining this with (76) controls the sum of the two terminal value-magnitude differences.

Finally, if $V, V' \in [0, B_v]$ are coupled nonnegative random variables, Tonelli's theorem gives

$$\begin{aligned} \int_0^{B_v} |P(V > y) - P(V' > y)| dy &\leq \mathbb{E} \int_0^{B_v} |\mathbf{1}\{V > y\} - \mathbf{1}\{V' > y\}| dy \\ &= \mathbb{E}|V - V'|. \end{aligned}$$

Since $|w_\sigma^\varepsilon(p) - w_\sigma^\varepsilon(q)| \leq C_1 |p - q|$, applying this inequality to the gain and loss integrals and using the preceding value bound proves (39). The decomposition (40) follows by adding and subtracting $J(P_{k+1}, z_{k+1}, r_k; \theta)$ and applying the triangle inequality uniformly in θ . $\square$

B.3 Score representation of the CPT gradient

Proof. We prove the gain part; the loss part follows by the same calculation and is subtracted at the end. To differentiate

$$J^+(\theta) = \int_0^{B_v} w_+^\varepsilon(S_\theta^+(y)) dy,$$

it is enough to identify $\nabla S_\theta^+(y)$ and justify moving the derivative through the y -integral.

For fixed y , the likelihood-ratio identity in Lemma A.1 applied to the bounded indicator gives

$$\nabla S_\theta^+(y) = \mathbb{E}\{\mathbf{1}[v_+(Y) > y] G_\theta\}. \quad (77)$$

By Cauchy-Schwarz and Assumption 3.1, the norm of the right-hand side is at most $\sqrt{C_{G,2}}$ uniformly in y and θ . Since $(w_+^\varepsilon)'$ is bounded by C_1 and the integration interval is finite, this constant is an integrable dominating function. The dominated convergence theorem therefore gives

$$\begin{aligned} \nabla J^+(\theta) &= \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \mathbb{E}\{\mathbf{1}[v_+(Y) > y] G_\theta\} dy \\ &= \mathbb{E} \left[G_\theta \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \mathbf{1}[v_+(Y) > y] dy \right], \end{aligned}$$

where Fubini's theorem is applicable because the integrand is dominated by $C_1 B_v \|G_\theta\|$ and $\mathbb{E}\|G_\theta\| < \infty$.

The score has mean zero. Therefore subtracting the deterministic quantity $S_\theta^+(y)$ inside the integral does not change the expectation. This yields the gain component of (41). Repeating the argument for the loss side and subtracting proves (42). The frozen-simulator identity follows under the same conditional argument with its trajectory law in place of the true law. $\square$

C Proofs for the Debiased Estimator

Throughout this appendix, condition on one frozen target and write $g = \nabla J(\theta)$. The complete trajectory summaries $\mathbf{X}_1, \mathbf{X}_2, \dots$ are independent and identically distributed under this conditional law. For a realization $x = (v^+, v^-, G_x)$, write $I_{x,\sigma}(y) = \mathbf{1}\{v^\sigma > y\}$.

C.1 Second-order expansion and full/half coupling

Proof of Lemma 5.4. To prove (43), it is enough to isolate the first-order Hoeffding projection and show that both the degenerate two-sample term and the Taylor remainder are $O_{L^2}(n^{-1})$. For a side σ and threshold y , write

$$S_\sigma(y) = P(V^\sigma > y), \quad e_{-i,\sigma}(y) = \widehat{S}_{-i,n}^\sigma(y) - S_\sigma(y).$$

Taylor's theorem applied to $(w_\sigma^\varepsilon)'$ gives

$$(w_\sigma^\varepsilon)'(S_\sigma + e) = (w_\sigma^\varepsilon)'(S_\sigma) + (w_\sigma^\varepsilon)''(S_\sigma)e + r_\sigma(e), \quad |r_\sigma(e)| \leq \frac{C_3}{2}e^2. \quad (78)$$

Substituting (78) into (30) splits the statistic into a first-order trajectory term, an ordered two-sample term, and a Taylor remainder.

Define

$$\begin{aligned} h_0(x) &= G_x \sum_\sigma \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(S_\sigma(y)) I_{x,\sigma}(y) dy, \\ \mathcal{K}(x, x') &= G_x \sum_\sigma \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)''(S_\sigma(y)) I_{x,\sigma}(y) \{I_{x',\sigma}(y) - S_\sigma(y)\} dy. \end{aligned}$$

By (42) and $\mathbb{E}G = 0$, $\mathbb{E}h_0(\mathbf{X}) = g$. Moreover $\mathbb{E}\{\mathcal{K}(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}\} = 0$ for independent $\mathbf{X}, \mathbf{X}'$, because $\mathbb{E}\{I_{\mathbf{X}',\sigma}(y) - S_\sigma(y)\} = 0$. Let

$$h_1(x') = \mathbb{E}_{\mathbf{X}}\{\mathcal{K}(\mathbf{X}, x')\}.$$

Then $\mathbb{E}h_1(\mathbf{X}') = 0$. Put $\mathcal{K}^\circ(x, x') = \mathcal{K}(x, x') - h_1(x')$. Both first-order projections of $\mathcal{K}^\circ$ vanish:

$$\mathbb{E}\{\mathcal{K}^\circ(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}\} = 0, \quad \mathbb{E}\{\mathcal{K}^\circ(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}'\} = 0.$$

The ordered two-sample term therefore has the exact decomposition

$$\frac{1}{n(n-1)} \sum_{i \neq j} \mathcal{K}(\mathbf{X}_i, \mathbf{X}_j) = \frac{1}{n} \sum_{j=1}^n h_1(\mathbf{X}_j) + \mathcal{D}_n, \quad (79)$$

where

$$\mathcal{D}_n = \frac{1}{n(n-1)} \sum_{i \neq j} \mathcal{K}^\circ(\mathbf{X}_i, \mathbf{X}_j).$$

The kernel is square-integrable because the indicators are bounded, the integration interval is finite, $(w_\sigma^\varepsilon)''$ is bounded, and $\mathbb{E}\|G\|_2^2 < \infty$. Degeneracy implies that covariances between two ordered kernel terms vanish unless their ordered index pairs coincide or are reversals of one another. Counting these $O(n^2)$ contributing pairs against the normalization $n^2(n-1)^2$ gives

$$\mathbb{E}\|\mathcal{D}_n\|_2^2 \leq \frac{C_D}{n^2} \quad (80)$$

for a finite constant C_D .

We next bound the Taylor remainder. For each i , $e_{-i,\sigma}$ is an empirical mean based on $n-1$ Bernoulli variables independent of $\mathbf{X}_i$. Uniformly in y ,

$$\mathbb{E}|e_{-i,\sigma}(y)|^4 \leq \frac{C_e}{(n-1)^2} \quad (81)$$

for a universal finite C_e . By (78), Cauchy–Schwarz in the y -integral, independence of G_i from the leave-one-out empirical survival, and $\mathbb{E}\|G_i\|_2^2 < \infty$, the i th Taylor contribution has second moment at most C_T/n^2 . Jensen’s inequality for the average over i then gives a second-moment bound of the same order for the averaged Taylor remainder T_n :

$$\mathbb{E}\|T_n\|_2^2 \leq \frac{C'_T}{n^2}. \quad (82)$$

Combining the first-order term, (79), and the Taylor remainder yields

$$U_n^{\text{raw}} - g = \frac{1}{n} \sum_{i=1}^n \mathfrak{h}(\mathbf{X}_i) + \mathcal{B}_n, \quad \mathfrak{h} = h_0 - g + h_1, \quad \mathcal{B}_n = \mathcal{D}_n + T_n.$$

The vector function $\mathfrak{h}$ has mean zero, and (80)–(82) imply $\mathbb{E}\|\mathcal{B}_n\|_2^2 \leq C_{\text{rem}}/n^2$ after increasing the finite constant C_{rem} . This proves (43).

Now take $n_\ell = n2^\ell$ and evaluate the full-sample statistic and its two half-sample statistics on the same n_ℓ trajectories. Applying (43) to the full sample and to each half gives

$$U_{n_\ell}^{\text{raw}} = g + \frac{1}{n_\ell} \sum_{i=1}^{n_\ell} \mathfrak{h}(\mathbf{X}_i) + \mathcal{B}_{n_\ell},$$

and

$$\frac{1}{2}\{U_{n_\ell/2}^{\text{raw},(1)} + U_{n_\ell/2}^{\text{raw},(2)}\} = g + \frac{1}{n_\ell} \sum_{i=1}^{n_\ell} \mathfrak{h}(\mathbf{X}_i) + \frac{1}{2}\{\mathcal{B}_{n_\ell/2}^{(1)} + \mathcal{B}_{n_\ell/2}^{(2)}\}.$$

The influence terms cancel exactly. Hence

$$D_{n,\ell}^{\text{raw}} = \mathcal{B}_{n_\ell} - \frac{1}{2}\{\mathcal{B}_{n_\ell/2}^{(1)} + \mathcal{B}_{n_\ell/2}^{(2)}\}.$$

Minkowski’s inequality in L^2 gives

$$(\mathbb{E}\|D_{n,\ell}^{\text{raw}}\|_2^2)^{1/2} \leq \frac{\sqrt{C_{\text{rem}}}}{n_\ell} + \frac{1}{2} \frac{2\sqrt{C_{\text{rem}}}}{n_\ell} + \frac{1}{2} \frac{2\sqrt{C_{\text{rem}}}}{n_\ell} = \frac{3\sqrt{C_{\text{rem}}}}{n_\ell}.$$

Squaring and substituting $n_\ell = n2^\ell$ proves (44).

It remains to prove (45). View U_n^{cen} as a symmetric function of $\mathbf{X}_1, \dots, \mathbf{X}_n$, and let $U_n^{\text{cen},(i)}$ be the statistic after replacing $\mathbf{X}_i$ by an independent copy $\mathbf{X}'_i$. If G'_i denotes the score component of $\mathbf{X}'_i$,

write $L_f = C_1 + C_2$. The coefficient in one centered LOO summand is bounded in absolute value by $2B_v C_1$, so the contribution from the replaced summand has norm at most

$$\frac{2B_v C_1}{n} (\|G_i\|_2 + \|G'_i\|_2).$$

For $j \neq i$, replacing $\mathbf{X}_i$ changes each leave-one-out survival entering the j th summand by at most $(n-1)^{-1}$ pointwise. Since

$$\left| \frac{\partial}{\partial p} \{ (w_\sigma^\varepsilon)'(p)(I-p) \} \right| \leq L_f,$$

the scalar coefficient multiplying G_j changes by at most $2B_v L_f / (n-1)$. Hence, for a deterministic constant C_* depending only on B_v, C_1, C_2 ,

$$\|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2 \leq \frac{C_*}{n} (\|G_i\|_2 + \|G'_i\|_2) + \frac{C_*}{n(n-1)} \sum_{j \neq i} \|G_j\|_2.$$

Using $(\sum_{j \neq i} x_j)^2 \leq (n-1) \sum_{j \neq i} x_j^2$ and the uniform second-moment bound for the score, there is a finite constant C_E such that

$$\mathbb{E} \|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2^2 \leq \frac{C_E}{n^2}$$

uniformly in i . The vector Efron–Stein inequality yields

$$\text{tr Cov}(U_n^{\text{cen}}) = \mathbb{E} \|U_n^{\text{cen}} - \mathbb{E} U_n^{\text{cen}}\|_2^2 \leq \frac{1}{2} \sum_{i=1}^n \mathbb{E} \|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2^2 \leq \frac{C_U}{n}$$

with $C_U = C_E/2$. This completes the proof. $\square$

C.2 Centering and finite-sample bias

Proof of Lemma 5.5. To compare the two LOO statistics, subtract (29) from (30). The difference is

$$U_n^{\text{raw}} - U_n^{\text{cen}} = \frac{1}{n} \sum_{i=1}^n G_i H_{-i}^{\text{ctr}}, \quad (83)$$

where

$$H_{-i}^{\text{ctr}} = \sum_{\sigma \in \{+, -\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) \hat{S}_{-i,n}^\sigma(y) dy.$$

The coefficient H_{-i}^{ctr} depends only on trajectories with indices different from i . It is therefore independent of G_i . The score identity gives $\mathbb{E} G_i = 0$, while Assumptions 3.1–3.2 make $G_i H_{-i}^{\text{ctr}}$ integrable. Hence every summand in (83) has mean zero and $\mathbb{E} U_n^{\text{cen}} = \mathbb{E} U_n^{\text{raw}} = \mu_n$.

It remains to control $\mu_n - g$. This follows from the expansion in Lemma 5.4. Taking expectations in (43) and using $\mathbb{E} \mathbf{h}(\mathbf{X}) = 0$ yields

$$\|\mu_n - g\|_2 = \|\mathbb{E} \mathcal{B}_n\|_2 \leq (\mathbb{E} \|\mathcal{B}_n\|_2^2)^{1/2} \leq \frac{\sqrt{C_{\text{rem}}}}{n}.$$

Thus (47) holds with $C_{\text{bias}} = \sqrt{C_{\text{rem}}}$. $\square$

C.3 Unbiasedness, variance, cost, and the CLT

Proof of Theorem 5.6. We first prove exact unbiasedness. By Lemma 5.5, the centered base has mean μ_n . For the raw correction,

$$\mathbb{E}D_{n,\ell}^{\text{raw}} = \mu_{n2^\ell} - \mu_{n2^{\ell-1}}. \quad (84)$$

The level bound (44) implies

$$\sum_{\ell \geq 1} \mathbb{E} \|D_{n,\ell}^{\text{raw}}\|_2 \leq \frac{3\sqrt{C_{\text{rem}}}}{n} \sum_{\ell \geq 1} 2^{-\ell} < \infty,$$

so the expectation can be interchanged with the infinite randomized series. Since (47) gives $\mu_{n2^{\ell_{\max}}} \rightarrow g$,

$$\begin{aligned} \mathbb{E}Z_n^{\text{db}} &= \mu_n + \sum_{\ell \geq 1} \mathbb{E}D_{n,\ell}^{\text{raw}} \\ &= \mu_n + \lim_{\ell_{\max} \rightarrow \infty} (\mu_{n2^{\ell_{\max}}} - \mu_n) = g. \end{aligned}$$

This proves (49).

For the second moment of the correction, independence of the activation and level variables gives

$$\begin{aligned} \mathbb{E} \left\| \frac{A}{\varrho \nu_{\ell^*}} D_{n,\ell^*}^{\text{raw}} \right\|_2^2 &= \frac{1}{\varrho} \sum_{\ell \geq 1} \frac{\mathbb{E} \|D_{n,\ell}^{\text{raw}}\|_2^2}{\nu_\ell} \\ &\leq \frac{9C_{\text{rem}}}{\varrho n^2} \sum_{\ell \geq 1} \frac{4^{-\ell}}{\nu_\ell} = \frac{9C_{\text{rem}}\Xi_b}{\varrho n^2}. \end{aligned}$$

The series Ξ_b is finite because $b < 2$. The base sample and correction sample are independent, so their covariance is zero. Combining the correction bound with (45) gives (50).

One base output costs exactly n complete trajectories. Conditional on activation and level ℓ , the correction uses one fresh batch of $n2^\ell$ trajectories; the two halves reuse this same batch. Hence

$$\mathbb{E} \text{cost}(Z_n^{\text{db}}) = n + \varrho n \sum_{\ell \geq 1} \nu_\ell 2^\ell = n(1 + \varrho \Upsilon_b),$$

where $\Upsilon_b < \infty$ because $b > 1$. Direct summation of the geometric series gives the closed forms in (48).

Finally, if $Z_{n,1}^{\text{db}}, \dots, Z_{n,M}^{\text{db}}$ are independent copies, exact unbiasedness and independence imply

$$\mathbb{E} \|\hat{g}_{M,n}^{\text{db}} - g\|_2^2 = \frac{1}{M} \text{tr Cov}(Z_n^{\text{db}}) \leq \frac{C_{\text{var}}(n)}{M}.$$

This is (52). □

Proof of Theorem 5.7. For fixed target and fixed (n, ϱ, b) , Theorem 5.6 gives iid vectors $Z_{n,r}^{\text{db}}$ with mean g and finite covariance Σ_n . For any fixed $t \in \mathbb{R}^d$, the scalar variables $t^\top (Z_{n,r}^{\text{db}} - g)$ are iid with finite variance. The classical scalar central limit theorem therefore gives

$$\sqrt{M} t^\top (\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, t^\top \Sigma_n t).$$

The Cramér–Wold device proves (53).

Because $\mathbb{E}\|Z_n^{\text{db}}\|_2^2 < \infty$, every entry of $Z_n^{\text{db}}(Z_n^{\text{db}})^\top$ is integrable. The law of large numbers therefore implies $\widehat{\Sigma}_{M,n} \rightarrow \Sigma_n$ in probability entrywise, hence in operator norm in fixed dimension. If Σ_n is nonsingular, operator-norm consistency implies $\mathbb{P}\{\lambda_{\min}(\widehat{\Sigma}_{M,n}) > \lambda_{\min}(\Sigma_n)/2\} \rightarrow 1$. On this event the inverse square-root map is well defined and continuous in a neighborhood of Σ_n . Slutsky’s theorem applied to (53) then proves (54). $\square$

D Proofs for the Online Residual and Tracking Results

D.1 Normalized projected ascent

Proof of Lemma 5.8. The target is a lower bound in terms of the population projected residual $Q(x)$. We first obtain a bound in terms of the actual displacement produced by $\widehat{g}$, then compare that displacement with the population displacement.

Let

$$s = x^+ - x, \quad c_g = \max\{\|g\|_2, \vartheta\}, \quad \mathfrak{e} = \mathbf{d}(\widehat{g}) - \mathbf{d}(g).$$

The projection variational inequality with the feasible comparison point x gives

$$\langle \mathbf{d}(\widehat{g}), s \rangle \geq \frac{\|s\|_2^2}{\gamma}. \quad (85)$$

Since $g = c_g \mathbf{d}(g)$, the L -smooth lower quadratic bound and (85) imply

$$\begin{aligned} J(x+s) - J(x) &\geq \langle g, s \rangle - \frac{L}{2} \|s\|_2^2 \\ &= c_g \langle \mathbf{d}(\widehat{g}) - \mathfrak{e}, s \rangle - \frac{L}{2} \|s\|_2^2 \\ &\geq \left(\frac{c_g}{\gamma} - \frac{L}{2} \right) \|s\|_2^2 - c_g \|\mathfrak{e}\|_2 \|s\|_2 \\ &\geq \left(\frac{3c_g}{4\gamma} - \frac{L}{2} \right) \|s\|_2^2 - c_g \gamma \|\mathfrak{e}\|_2^2 \\ &\geq \frac{c_g}{4\gamma} \|s\|_2^2 - c_g \gamma \|\mathfrak{e}\|_2^2. \end{aligned} \quad (86)$$

The penultimate line uses $uv \leq u^2/(4\gamma) + \gamma v^2$, and the last line uses $\gamma L \leq \vartheta \leq c_g$.

Now let

$$t = \Pi_{\Theta}\{x + \gamma \mathbf{d}(g)\} - x = \gamma Q(x).$$

Nonexpansiveness of Euclidean projection gives $\|s - t\|_2 \leq \gamma \|\mathfrak{e}\|_2$. From $\|t\|_2^2 \leq 2\|s\|_2^2 + 2\|s - t\|_2^2$,

$$\|s\|_2^2 \geq \frac{1}{2} \|t\|_2^2 - \gamma^2 \|\mathfrak{e}\|_2^2.$$

Substituting into (86) yields

$$J(x^+) - J(x) \geq \frac{c_g \gamma}{8} \|Q(x)\|_2^2 - \frac{5c_g \gamma}{4} \|\mathfrak{e}\|_2^2.$$

Finally, $\mathbf{d}(g) = \Pi_{\{u: \|u\|_2 \leq 1\}}(g/\vartheta)$, so nonexpansiveness of the unit-ball projection implies

$$\|\mathfrak{e}\|_2 \leq \vartheta^{-1} \|\widehat{g} - g\|_2.$$

Using $c_g \geq \vartheta$ in the positive term and $c_g \leq C_g = \max(B_g, \vartheta)$ in the negative term proves (55). $\square$

D.2 Average expected squared projected residual

Proof of Theorem 5.9. We first connect the final estimator to the true gradient. Conditional on $\mathcal{F}_k$, Theorem 5.6 gives

$$\mathbb{E}(\hat{g}_k^{\text{db}} - \tilde{g}_k(\theta_k) \mid \mathcal{F}_k) = 0, \quad \mathbb{E}(\|\hat{g}_k^{\text{db}} - \tilde{g}_k(\theta_k)\|_2^2 \mid \mathcal{F}_k) \leq \frac{C_{\text{var}}(n)}{M_k}.$$

The cross term with the deterministic conditional vector $\tilde{g}_k(\theta_k) - g_k(\theta_k)$ therefore vanishes. Hence

$$\begin{aligned} \mathbb{E}\|\hat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2 &= \mathbb{E}\|\hat{g}_k^{\text{db}} - \tilde{g}_k(\theta_k)\|_2^2 + \mathbb{E}\|\tilde{g}_k(\theta_k) - g_k(\theta_k)\|_2^2 \\ &\leq \frac{C_{\text{var}}(n)}{M_k} + \mathbb{E}\beta_k^2, \end{aligned}$$

which is (56).

We next apply Lemma 5.8 to the true episode objective J_k at $x = \theta_k$ and $\hat{g} = \hat{g}_k^{\text{db}}$. Let

$$\mathbf{j}_k = J_k(\theta_{k+1}) - J_k(\theta_k).$$

After taking expectations,

$$\frac{\vartheta\gamma}{8}\mathbb{E}\zeta_k \leq \mathbb{E}\mathbf{j}_k + \frac{5C_g\gamma}{4\vartheta^2}\mathbb{E}\|\hat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2. \quad (87)$$

The objective increments telescope across moving objectives as the pathwise identity

$$\begin{aligned} \sum_{k=1}^K \mathbf{j}_k &= J_K(\theta_{K+1}) - J_1(\theta_1) \\ &\quad + \sum_{k=1}^{K-1} \{J_k(\theta_{k+1}) - J_{k+1}(\theta_{k+1})\}. \end{aligned} \quad (88)$$

Since $|J_k| \leq B_v$ and the second line is bounded above by $\mathcal{V}_K$,

$$\sum_{k=1}^K \mathbf{j}_k \leq 2B_v + \mathcal{V}_K. \quad (89)$$

Summing (87), applying (56) and (89), and dividing by $\vartheta\gamma K/8$ gives exactly (57). $\square$

D.3 Discounted dynamic local regret

Proof of Corollary 5.10. Write $W = W_{\rho,w}$. With the convention $Q_t(\theta_t) = 0$ for $t \leq 0$, the coefficients ρ^j/W , $j = 0, \dots, w-1$, are nonnegative and sum to one. Convexity of $q \mapsto \|q\|_2^2$ therefore gives

$$\|\bar{Q}_k\|_2^2 \leq \frac{1}{W} \sum_{j=0}^{w-1} \rho^j \|Q_{k-j}(\theta_{k-j})\|_2^2.$$

Summing over $k \leq K$ and reindexing,

$$\mathcal{R}_{\rho,w}(K) \leq \sum_{t=1}^K \zeta_t \left\{ \frac{1}{W} \sum_{j=0}^{\min\{w-1, K-t\}} \rho^j \right\} \leq \sum_{t=1}^K \zeta_t,$$

which proves (58). Taking expectations and dividing by K gives (59). $\square$

D.4 Stationary-set consequences

Proof of Corollary 5.11. Assumption 3.4 gives pointwise

$$\text{dist}^2(\theta_k, \mathbf{S}_k) \leq \kappa^2 \|Q_k(\theta_k)\|_2^2 = \kappa^2 \zeta_k.$$

Taking expectations and averaging over k proves (60). Substitution of (57) yields the stated finite-horizon control. $\square$

Proof of Lemma 5.12. The normalized direction map is ϑ^{-1} -Lipschitz because it is the Euclidean projection of g/ϑ onto the closed unit ball. Using nonexpansiveness of Π_Θ in (15),

$$\begin{aligned} \sup_{\theta \in \Theta} \|Q_{k+1}(\theta) - Q_k(\theta)\|_2 &\leq \sup_{\theta \in \Theta} \|\mathbf{d}(g_{k+1}(\theta)) - \mathbf{d}(g_k(\theta))\|_2 \\ &\leq \frac{\delta_k^g}{\vartheta}. \end{aligned}$$

If $\theta \in \mathbf{S}_k$, then $Q_k(\theta) = 0$, so Assumption 3.4 applied to episode $k+1$ gives

$$\text{dist}(\theta, \mathbf{S}_{k+1}) \leq \kappa \|Q_{k+1}(\theta)\|_2 \leq \frac{\kappa \delta_k^g}{\vartheta}.$$

Taking the supremum over $\theta \in \mathbf{S}_k$ gives one directed Hausdorff bound. Interchanging k and $k+1$ gives the reverse bound because the same uniform gradient-difference bound is symmetric. Taking the maximum proves (62). $\square$

Proof of Corollary 5.13. In (57), the endpoint term is $O(K^{-1})$. The assumption $\mathbb{E}\mathcal{V}_K = o(K)$ makes the variation term vanish after division by K . Since $C_{\text{var}}(n)$ is fixed,

$$\frac{1}{K} \sum_{k=1}^K \left(\frac{C_{\text{var}}(n)}{M_k} + \mathbb{E}\beta_k^2 \right) \rightarrow 0$$

by the assumed Cesàro condition. Therefore $\mathcal{E}_K \rightarrow 0$. Equation (59) gives the DLR conclusion, and (60) gives the stationary-set conclusion when Assumption 3.4 is imposed. $\square$

E Proof of the Fixed-Dimension Minimax Rate

Proof of Theorem 5.14. We prove the lower and upper bounds under the same complete-trajectory oracle and the same expected trajectory budget. For the lower bound, it is enough to construct a two-point submodel contained in $\mathfrak{C}$ whose CPT gradients are separated by order $\mathbf{B}^{-1/2}$ while the stopped transcript distributions remain at bounded total variation. For the upper bound, Theorem 5.6 supplies an unbiased finite-variance output with finite expected trajectory cost.

Step 1: a CPT-realizable one-stock submodel. Consider one risky asset and cash, horizon $h = 1$, zero transaction cost, and episode-start wealth and reference equal to one. Take the previous portfolio to be all cash and let the feasibility map be the identity. Set the risky feature to $e_1 \in \mathbb{R}^d$ and the cash feature to zero. At a query θ , the latent risky weight satisfies

$$\mathbf{U} \sim \text{Beta}(e^{\theta^{[1]}}, 1),$$

where $\theta^{[1]}$ is the first coordinate of θ . After the inertia map (7), the executed risky weight is $\chi \mathbf{U}$.

Let the risky return be $R = R_\star \xi$, where $R_\star > 0$ is fixed and sufficiently small and $\xi \sim \text{Bernoulli}(p)$ is independent of $\mathbf{U}$. The terminal wealth and reference are

$$X_{\text{end}} = 1 + \chi R_\star \mathbf{U} \xi, \quad r_{\text{end}} = 1 + \eta_+ \chi R_\star \mathbf{U} \xi,$$

so the terminal relative outcome is

$$Y = v \mathbf{U} \xi, \quad v = (1 - \eta_+) \chi R_\star > 0. \quad (90)$$

This is an admissible portfolio trajectory model under the common bounds in Assumption 3.6. The loss-side value is identically zero.

Step 2: local separation of the CPT gradient. Let $v = v_+$ and define

$$\mathbf{m}(t) = v v'(vt), \quad 0 \leq t \leq 1.$$

The shifted-power value function gives $\mathbf{m}(0) = 0$, $\mathbf{m}(t) > 0$ for $t \in (0, 1]$, and bounded $\mathbf{m}$. Since $P(\mathbf{U} > t) = 1 - t^{e^{\theta^{[1]}}}$, a change of variables $y = v(vt)$ in the gain-side CPT integral gives

$$J_p(\theta) = \int_0^1 \mathbf{m}(t) w_+^\varepsilon \left(p \{1 - t^{e^{\theta^{[1]}}}\} \right) dt. \quad (91)$$

At the target $\theta_0 = 0$, differentiation under the integral is valid because, on a compact neighborhood of θ_0 , $t^{e^{\theta^{[1]}}} |\log t|$ has an integrable envelope on $(0, 1)$ and $(w_+^\varepsilon)'$ is bounded. Therefore

$$\nabla J_p(0) = \mathbf{g}(p) e_1, \quad \mathbf{g}(p) = p \int_0^1 \mathbf{m}(t) (w_+^\varepsilon)'(p(1-t)) (-t \log t) dt. \quad (92)$$

Put

$$A_v = \int_0^1 \mathbf{m}(t) (-t \log t) dt > 0, \quad d_W = (w_+^\varepsilon)'(0) > 0.$$

Since $|(w_+^\varepsilon)''| \leq C_2$, differentiation of (92) with respect to p gives

$$\begin{aligned} \mathbf{g}'(p) &= \int_0^1 \mathbf{m}(t) (-t \log t) \left[(w_+^\varepsilon)'(p(1-t)) + p(1-t) (w_+^\varepsilon)''(p(1-t)) \right] dt \\ &\geq (d_W - 2C_2 p) A_v. \end{aligned}$$

Choose $p_{\max} \in (0, 1/2]$ small enough that $2C_2 p_{\max} \leq d_W/2$; if $C_2 = 0$, any choice in $(0, 1/2]$ is admissible. Let

$$p_* = p_{\max}/2, \quad \mathcal{I} = [p_*/2, 3p_*/2], \quad c_* = d_W A_v/2.$$

Then

$$\mathbf{g}'(p) \geq c_* > 0 \quad \text{for every } p \in \mathcal{I}. \quad (93)$$

Step 3: information in an adaptive stopped transcript. For B large enough, take

$$p_\pm = p_* \pm \delta, \quad \delta = \delta_*/\sqrt{B},$$

where $\delta_* > 0$ is chosen so that $p_\pm \in \mathcal{I}$. For Bernoulli laws in the fixed interior interval $\mathcal{I}$, the elementary bound $\text{kl}(p||q) \leq (p-q)^2/\{q(1-q)\}$ gives

$$\text{kl}(p_+||p_-) \leq C_I \delta^2 \quad (94)$$

for a finite constant C_I depending only on $\mathcal{I}$.

Consider an arbitrary admissible estimator in (67). Its query point at each oracle call may depend on the preceding transcript. Conditional on any such query, the latent Dirichlet action law is identical under p_+ and p_- ; only the Bernoulli return kernel differs. All algorithmic randomization can be included in the transcript and is likewise common to the two models. The data-dependent call count T_{call} is a stopping time for the oracle transcript filtration. The chain rule for the stopped likelihood ratio gives

$$\text{KL}(\mathbb{P}_+^{\text{tr}} \parallel \mathbb{P}_-^{\text{tr}}) = \mathbb{E}_+ T_{\text{call}} \text{kl}(p_+ \parallel p_-) \leq \mathbf{B} C_I \delta^2 = C_I \delta_*^2. \quad (95)$$

The equality can be obtained by summing the conditional log-likelihood increments over $\{T_{\text{call}} \geq j\}$; these events are measurable before the j th fresh return is observed. The increments are uniformly integrable because $p_{\pm}$ stay in the compact interior interval $\mathcal{I}$ and $\mathbb{E}_+ T_{\text{call}} < \infty$.

Choose δ_* small enough that $C_I \delta_*^2 \leq 1/8$. Pinsker's inequality then gives

$$\text{TV}(\mathbb{P}_+^{\text{tr}}, \mathbb{P}_-^{\text{tr}}) \leq \frac{1}{4}. \quad (96)$$

Step 4: reduction from estimation to testing. By (93) and the mean-value theorem, the two gradient targets satisfy

$$D_{\mathbf{B}}^g := \|\mathbf{g}(p_+)e_1 - \mathbf{g}(p_-)e_1\|_2 \geq 2c_*\delta. \quad (97)$$

Given any estimator $\hat{g}$, define the test that chooses the closer of the two targets. On a testing error, the estimator must be at least $D_{\mathbf{B}}^g/2$ from the true target. The sum of the two testing error probabilities is at least one minus the total variation distance. Therefore

$$\begin{aligned} \max_{s \in \{+, -\}} \mathbb{E}_s \|\hat{g} - \mathbf{g}(p_s)e_1\|_2^2 &\geq \frac{(D_{\mathbf{B}}^g)^2}{8} \{1 - \text{TV}(\mathbb{P}_+^{\text{tr}}, \mathbb{P}_-^{\text{tr}})\} \\ &\geq \frac{3c_*^2\delta_*^2}{8\mathbf{B}}. \end{aligned}$$

The two models belong to $\mathfrak{C}$, so taking the infimum over all admissible procedures gives the lower bound in (68) with $c_{\text{lb}} = 3c_*^2\delta_*^2/8$.

Step 5: cost-matched upper bound. Fix any admissible estimator design parameters (n, ϱ, b) from Theorem 5.6. The common model-class constants in Assumption 3.6 make

$$\sup_{P \in \mathfrak{C}} \text{tr Cov}_P(Z_n^{\text{db}}) \leq C_{\text{var}}(n) < \infty, \quad C_{\text{cost}} = n(1 + \varrho\Upsilon_b) < \infty$$

uniform over $P \in \mathfrak{C}$. For

$$M = \left\lfloor \frac{\mathbf{B}}{C_{\text{cost}}} \right\rfloor,$$

run M independent, untruncated debiased outputs Z_n^{db} at θ_0 and average them. Their total expected number of complete trajectories is $MC_{\text{cost}} \leq \mathbf{B}$, so this is an admissible procedure. For $\mathbf{B} \geq 2C_{\text{cost}}$, $M \geq \mathbf{B}/(2C_{\text{cost}})$. Hence

$$\sup_{P \in \mathfrak{C}} \mathbb{E}_P \|\hat{g}_{M,n}^{\text{db}} - g_P\|_2^2 \leq \frac{C_{\text{var}}(n)}{M} \leq \frac{2C_{\text{var}}(n)C_{\text{cost}}}{\mathbf{B}}.$$

Thus the upper bound in (68) holds with $C_{\text{ub}} = 2C_{\text{var}}(n)C_{\text{cost}}$ and a sufficiently large $\mathbf{B}_0$. The lower and upper bounds use the same target, trajectory oracle, loss, and expected raw-trajectory budget, completing the proof. $\square$